

Signal and System Design (IGS2137)

Jump Together, Fly Farther! ***Weeks 6&7 Lectures***



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Chapter 3

Fourier Representation for Signal and LTIV Systems



3.1 Introductions

- Represent a signal as a weighted superposition of complex sinusoids.
- The study of sinusoidal representation is termed Fourier analysis after Joseph Fourier (1768-1830) for his contributions to the theory of representing functions as weighted superposition of sinusoids.



3.2 Complex Sinusoids and Frequency Response of LTI Systems

- Frequency response : the response of an LTI system to a sinusoidal input, $x[n]=e^{j\Omega n}$
- A complex sinusoid input to a LTI system generates an output equal to the sinusoidal input multiplied by the system frequency response



3.2 Complex Sinusoids and Frequency Response of LTI Systems

■ Complex Sinusoids and LTI Systems

- $$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$
$$= \sum_{k=-\infty}^{\infty} h[k]e^{j\Omega(n-k)}, \quad \text{for complex sinusoidal signal, } x[n] = e^{j\Omega n}$$
$$= e^{j\Omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$$
$$= H(e^{j\Omega})e^{j\Omega n},$$

where $H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$: frequency response of D.T. system

- $$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{j\omega(t-\tau)}d\tau = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$$
$$= H(j\omega)e^{j\omega t},$$

where $H(j\omega) = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$: frequency response of C.T. system



3.2 Complex Sinusoids and Frequency Response of LTI Systems

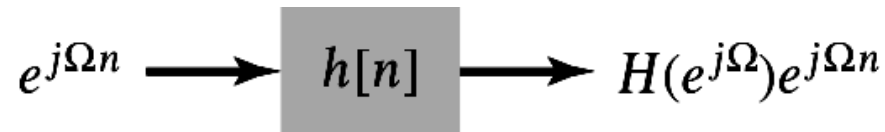


Figure 3.1 (p. 196)

The output of a complex sinusoidal input to an LTI system is a complex sinusoid of the same frequency as the input, multiplied by the frequency response of the system.



$$y[n] = e^{j\Omega n} \sum_{k=-\infty}^{\infty} h[k] e^{-j\Omega k} = H(e^{j\Omega}) e^{j\Omega n}$$

Complex scaling factor

3. Frequency response:

$$H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\Omega k} \quad (3.1)$$

A function of frequency Ω

♣ Continuous-time LTI system

1. Impulse response of continuous-time LTI system = $h(t)$, input = $x(t) = e^{j\omega t}$

2. Output:

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \\ &= H(j\omega) e^{j\omega t} \end{aligned} \quad (3.2)$$



3.2 Complex Sinusoids and Frequency Response of LTI Systems

3. Frequency response:

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \quad (3.3)$$

♣ Polar form complex number $c = a + jb$:

$$c = |c| e^{j \arg\{c\}} \quad \text{where} \quad |c| = \sqrt{a^2 + b^2} \quad \text{and} \quad \arg\{c\} = \tan^{-1}\left(\frac{b}{a}\right)$$

♣ Polar form for $H(j\omega)$:

$$H(j\omega) = |H(j\omega)| e^{j \arg\{H(j\omega)\}}$$

where $|H(j\omega)|$ = Magnitude response and $\arg\{H(j\omega)\}$ = Phase response



$$y(t) = |H(j\omega)| e^{j(\omega t + \arg\{H(j\omega)\})}$$



3.2 Complex Sinusoids and Frequency Response of LTI Systems

Example 3.1 *RC Circuit: Frequency response*

The impulse response of the system relating to the input voltage to the voltage across the capacitor in **Fig. 3.2** is derived in **Example 1.21** as

$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

Find an expression for the frequency response, and plot the magnitude and phase response.



3.2 Complex Sinusoids and Frequency Response of LTI Systems

<Sol.> Frequency response:

$$H(j\omega) = \frac{1}{RC} \int_{-\infty}^{\infty} e^{-\frac{\tau}{RC}} u(\tau) e^{-j\omega\tau} d\tau = \frac{1}{RC} \int_0^{\infty} e^{-\left(j\omega + \frac{1}{RC}\right)\tau} d\tau$$

$$= \frac{1}{RC} \left[\frac{-1}{\left(j\omega + \frac{1}{RC}\right)} e^{-\left(j\omega + \frac{1}{RC}\right)\tau} \right]_0^{\infty}$$

$$= \frac{1}{RC} \frac{-1}{\left(j\omega + \frac{1}{RC}\right)} (0 - 1)$$

$$= \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$

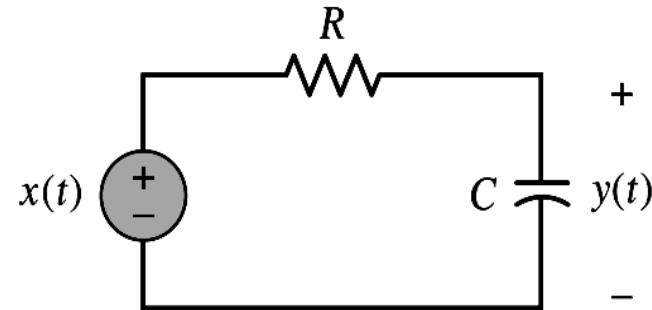


Figure 3.2 (p. 197)
RC circuit for Example 3.1.

Magnitude response:

$$|H(j\omega)| = \frac{\frac{1}{RC}}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$$

Phase response:

$$\arg\{H(j\omega)\} = -\arctan(\omega RC)$$



Fig. 3.6 (a) and (b)



3.2 Complex Sinusoids and Frequency Response of LTI Systems

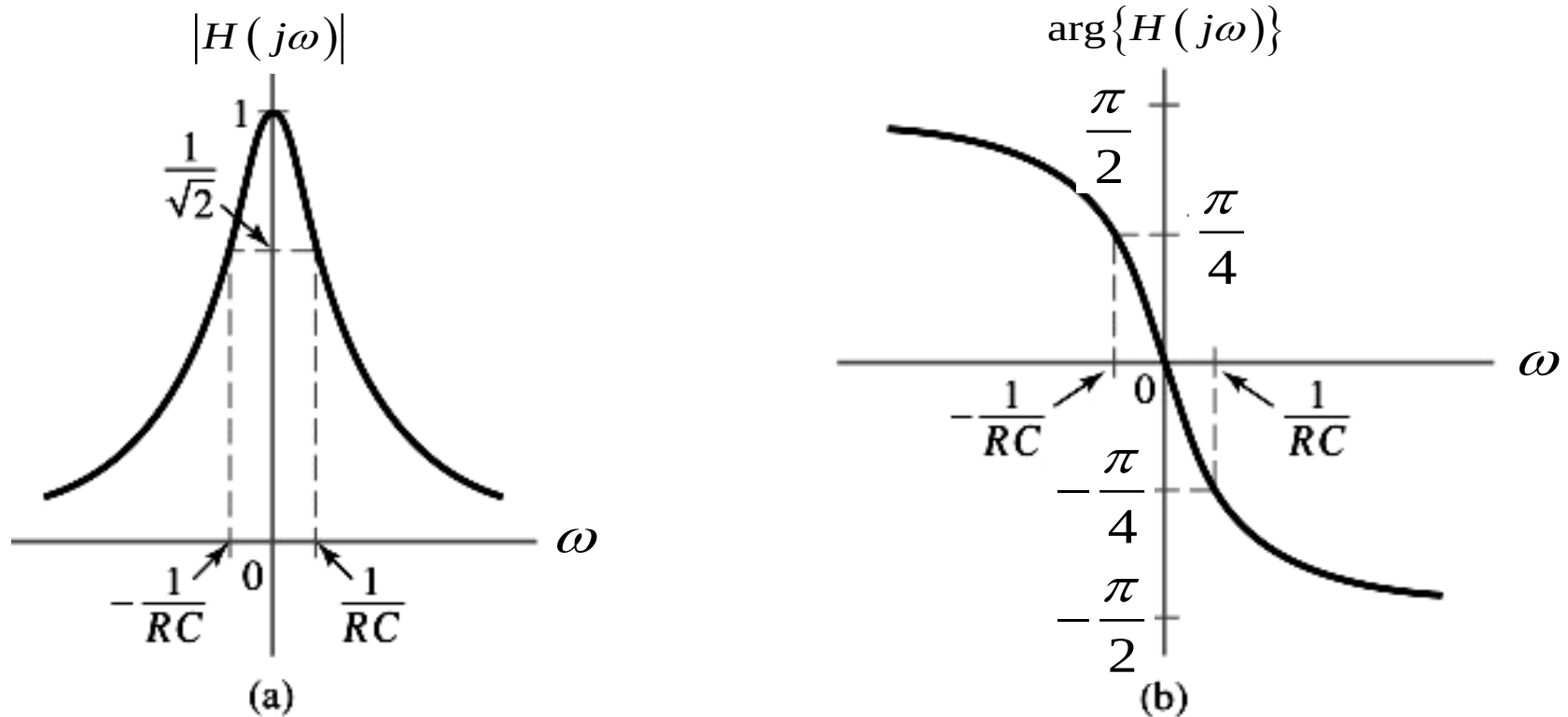


Figure 3.3 (p. 198)

Frequency response of the RC circuit in Fig. 3.2.

(a) Magnitude response. (b) Phase response.

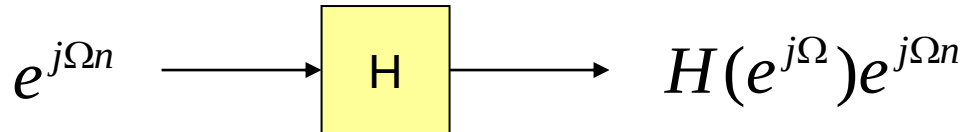
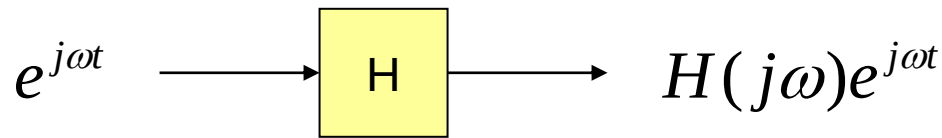
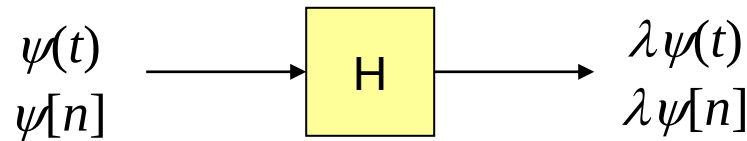


3.2 Complex Sinusoids and Frequency Response of LTI Systems

Complex Sinusoids and LTI Systems

- Eigenfunction and eigenvalue of a system

Eigenfunction: $\psi(t) = e^{j\omega t}$ Eigenvalue: $\lambda = H(j\omega)$



$$x(t) = \sum_{k=1}^M a_k e^{j\omega_k t} \rightarrow y(t) = \sum_{k=1}^M a_k H(j\omega_k) e^{j\omega_k t}$$



3.2 Complex Sinusoids and Frequency Response of LTI Systems

■ Complex Sinusoids and LTI Systems

- Eigenfunction and eigenvalue of a system

Eigenfunction: $\psi(t) = e^{j\omega t}$ Eigenvalue: $\lambda = H(j\omega)$

If $\mathbf{e}_k$ is an eigenvector of a matrix $\mathbf{A}$ with eigenvalue λ_k , then

$$\mathbf{A}\mathbf{e}_k = \lambda_k \mathbf{e}_k$$

Arbitrary input = weighted superpositions of eigenfunctions

⇒ Convolution operation ⇒ Multiplication

Ex. Input:

$$x(t) = \sum_{k=1}^M a_k e^{j\omega_k t}$$

Output:

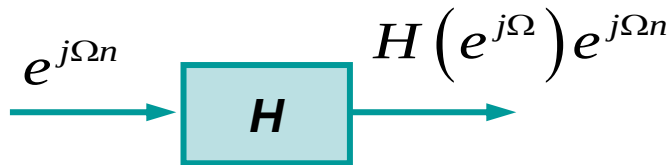
$$y(t) = \sum_{k=1}^M a_k H(j\omega) e^{j\omega_k t}$$



3.2 Complex Sinusoids and Frequency Response of LTI Systems

Discrete-time case:

1. Eigenrepresentation: Fig. 3.4 (c) $\Rightarrow H(\psi[n]) = \lambda \psi[n]$



Eigenfunction: $\psi[n] = e^{j\Omega n}$

Eigenvalue: $\lambda = H(e^{j\Omega})$

2. Input:

$$x[n] = \sum_{k=1}^M a_k e^{j\Omega_k n}$$



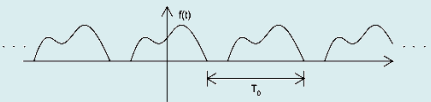
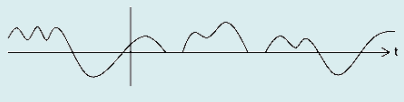
Output:

$$y[n] = \sum_{k=1}^M a_k H(e^{j\Omega_k}) e^{j\Omega_k n}$$



3.3 Fourier Representation for Four Classes of Signals

Table 3.1 Relationship between Time Properties of a Signal and the Approximate Fourier Representation

Time Frequency	 Periodic	 Non-periodic
Continuous (t)	Fourier Series (FS)	Fourier Transform (FT)
Discrete [n]	Discrete -Time Fourier Series (DTFS)	Discrete-Time Fourier Transform (DTFT)

Note: frequency representation of D.T. signal does not generate discrete Freq.



3.2 Complex Sinusoids and Frequency Response of LTI Systems

3.3.1 Periodic Signals: Fourier Series Representations

1. $x[n]$ = discrete-time signal with fundamental period N . DTFS of $x[n]$ is

$$\hat{x}[n] = \sum_k A[k] e^{jk\Omega_0 n}$$

(3.4)

$\Omega_0 = 2\pi/N \equiv$ Fundamental frequency of $x[n]$

2. $x(t)$ = continuous-time signal with fundamental period T . FS of $x(t)$ is

$$\hat{x}(t) = \sum_k A[k] e^{jk\omega_0 t}$$

(3.5)

$\omega_0 = 2\pi/T \equiv$ Fundamental frequency of $x(t)$

♣ $e^{jk\omega_0 t}$ is the k th harmonic of $e^{j\omega_0 t}$.



3.2 Complex Sinusoids and Frequency Response of LTI Systems

3. The complex sinusoids $\exp(jk\Omega_0 n)$ are N -periodics in the frequency index k .

<pf.>
$$e^{j(N+K)\Omega_0 n} = e^{jN\Omega_0 n} e^{jK\Omega_0 n} = e^{j2\pi m} e^{jN\Omega_0 n} = e^{jK\Omega_0 n}$$

⇒ There are only N distinct complex sinusoids of the form $\exp(jk\Omega_0 n)$ s should be used in Eq. (3.4).

Eq. (3.4)



$$\hat{x}(t) = \sum_{k=0}^{N-1} A[k] e^{jk\Omega_0 n} \quad (3.6)$$

♣ Symmetries in $x[k]$: $k = -(N-1)/2$ to $(N-1)/2$

4. Continuous-time complex sinusoid $\exp(jk\omega_0 t)$ with distinct frequencies $k\omega_0$ are always distinct.

FS of continuous-time periodic signal $x(t)$ becomes

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} A[k] e^{jk\omega_0 t} \quad (3.7)$$



3.3 Fourier Representation for Four Classes of Signals

■ Basic Theory of Inner Product Space

*Functional
Space*

- Inner Product: $\langle f, g \rangle = \int_{\langle t \rangle} f(t) g^*(t) dt$
- Basis Functions: $\{f_1, f_2, f_3, \dots\}$
- $f (\in \text{Function Space}) = \sum_i c_i f_i$
- Orthonormality: $\langle f_i, f_j \rangle = \delta(i - j)$
- The coefficient $c_i : c_i = \langle f, f_i \rangle$



3.3 Fourier Representation for Four Classes of Signals

■ Orthogonality of Complex Sinusoids

- The orthogonality of complex sinusoids plays a key role in Fourier representations.
- Orthogonal if its inner product is zero.

- Discrete time case

$$\Omega_0 = \frac{2\pi}{N}$$

$$I_{k,m} = \sum_{n=0}^{N-1} e^{j(k-m)\Omega_0 n} = \begin{cases} N, & k = m \\ \frac{1 - e^{j(k-m)2\pi}}{1 - e^{j(k-m)\Omega_0}}, & k \neq m \end{cases} = \begin{cases} N, & k = m \\ 0, & k \neq m \end{cases}$$

- Continuous time case

$$I_{k,m} = \int_0^T e^{j(k-m)\omega_0 t} dt = \begin{cases} T, & k = m \\ \frac{1}{j(k-m)\omega_0} e^{j(k-m)\omega_0 t} \Big|_0^T, & k \neq m \end{cases} = \begin{cases} T, & k = m \\ 0, & k \neq m \end{cases}$$



3.4 D.T. Periodic Signals: D.T. Fourier Series

- The DTFS represents an N periodic($T=N$) discrete-time signal $x[n]$ as the series of

$$x[n] = \sum_{k=\langle N \rangle} A[k] e^{jk\Omega_0 n}, \quad \text{where } \Omega_0 = \frac{2\pi}{N}$$

- MSE problem

$$\begin{aligned} MSE &= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n] - \hat{x}[n]|^2 \\ &= \frac{1}{N} \sum_{n=\langle N \rangle} \left| x[n] - \sum_{k=\langle N \rangle} A[k] e^{jk\Omega_0 n} \right|^2 \end{aligned}$$

Then, $\min_{A[k]} MSE$



3.4 D.T. Periodic Signals: D.T. Fourier Series

$$\begin{aligned}
 MSE &= \frac{1}{N} \sum_{n=\langle N \rangle} \left[\left(x[n] - \sum_{k=\langle N \rangle} A[k] e^{jk\Omega_0 n} \right) \left(x[n] - \sum_{m=\langle N \rangle} A[m] e^{jm\Omega_0 n} \right)^* \right] \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 - \sum_{m=\langle N \rangle} A^*[m] \left(\frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jm\Omega_0 n} \right) - \sum_{k=\langle N \rangle} A[k] \left(\frac{1}{N} \sum_{n=\langle N \rangle} x^*[n] e^{jk\Omega_0 n} \right) \\
 &\quad + \sum_{k=\langle N \rangle} \sum_{m=\langle N \rangle} A^*[m] A[k] \left(\frac{1}{N} \sum_{n=\langle N \rangle} e^{j(k-m)\Omega_0 n} \right)
 \end{aligned}$$

$$\text{Let } X[k] = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\Omega_0 n}$$

$$\begin{aligned}
 \therefore MSE &= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 - \sum_{k=\langle N \rangle} A^*[k] X[k] - \sum_{k=\langle N \rangle} A[k] X^*[k] + \sum_{k=\langle N \rangle} |A[k]|^2 \\
 &= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 + \sum_{k=\langle N \rangle} |A[k] - X[k]|^2 - \sum_{k=\langle N \rangle} |X[k]|^2
 \end{aligned}$$

To minimise MSE, $A[k] = X[k]$

$$\text{Then, } MSE = \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 - \sum_{k=\langle N \rangle} |X[k]|^2$$



3.4 D.T. Periodic Signals: D.T. Fourier Series

- The value of the minimum MSE

$$\begin{aligned}\sum_{k=\langle N \rangle} |X[k]|^2 &= \sum_{k=\langle N \rangle} \left(\frac{1}{N^2} \sum_{n=\langle N \rangle} \sum_{m=\langle N \rangle} x[n] x^*[m] e^{j(m-n)\Omega_0 k} \right) \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{m=\langle N \rangle} x[n] x^*[m] \frac{1}{N} \sum_{k=\langle N \rangle} e^{j(m-n)\Omega_0 k} \\&\quad \left(\because \text{Orthogonality: } \frac{1}{N} \sum_{k=\langle N \rangle} e^{j(m-n)\Omega_0 k} = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases} \right) \\&= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 \\ \therefore MSE &= \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 - \sum_{k=\langle N \rangle} |X[k]|^2 = 0 \\ \therefore \frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 &= \sum_{k=\langle N \rangle} |X[k]|^2\end{aligned}$$



3.4 D.T. Periodic Signals: D.T. Fourier Series

- $X[k]$ is N periodic in k

$$\begin{aligned} X[N + k] &= \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-j(k+N)\Omega_0 n} \\ &= \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-jk\Omega_0 n} e^{-jN\Omega_0 n} \\ (e^{-jN\Omega_0 n} &= e^{-j2\pi n}, \quad \Omega_0 = 2\pi/N) \\ &= \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-jk\Omega_0 n} = X[k] \end{aligned}$$



3.4 D.T. Periodic Signals: D.T. Fourier Series

- The DTFS representation for $x[n]$ is given by

$$\begin{aligned} \text{Synthesis : } x[n] &= \sum_{k=\langle N \rangle} X[k] e^{jk\Omega_0 n} \\ \text{Analysis : } X[k] &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\Omega_0 n} \end{aligned}$$

where $x[n]$ has a fundamental period N and $\Omega_0 = 2\pi/N$.

$$x[n] \overset{\text{DTFS}; \Omega_0}{\longleftrightarrow} X[k]$$



3.4 D.T. Periodic Signals: D.T. Fourier Series

Ex) Define a partial sum approximation to $x[n]$ as

$$B[k] = \begin{cases} X[k], & k = 0, N/2 \\ 2X[k], & k = 1, 2, \dots, N/2 - 1 \end{cases}$$

$$x_J[n] = \sum_{k=0}^J B[k] \cos(k\Omega_0 n) \quad (3.18)$$

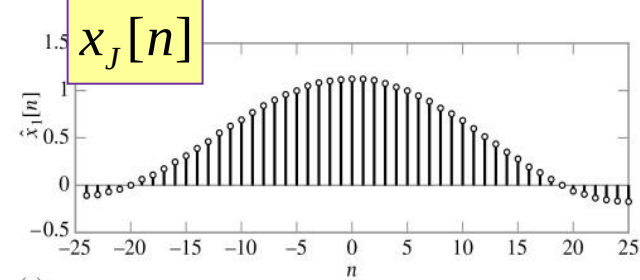
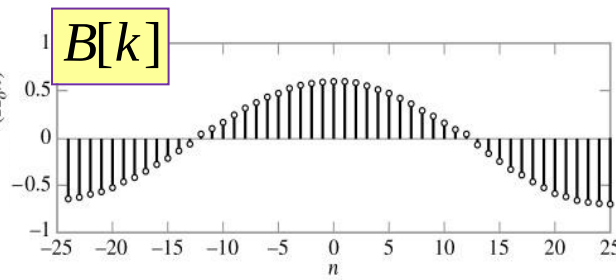
where $J \leq N/2$. $N=50$ and $M=12$, $J=1, 3, 5, 23$, and 25.



$$B[k] = \begin{cases} X[k], k = 0, N/2 \\ 2X[k], k = 1, 2, \dots, N/2 - 1 \end{cases}$$

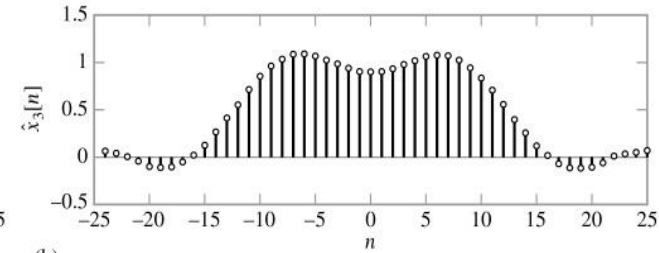
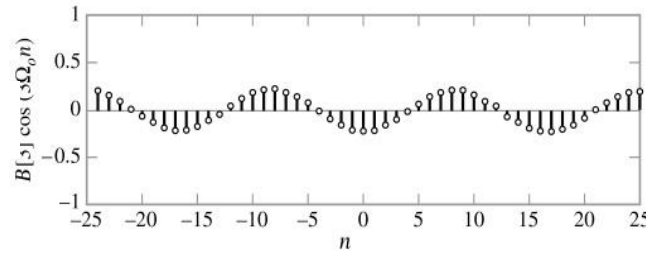
$$x_J[n] = \sum_{k=0}^J B[k] \cos(k\Omega_0 n)$$

$J=1$



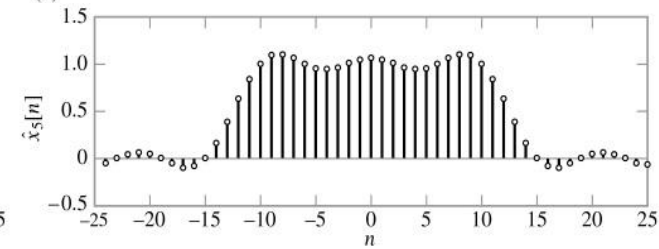
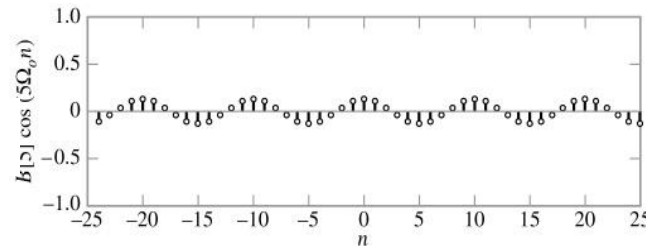
(a)

$J=3$



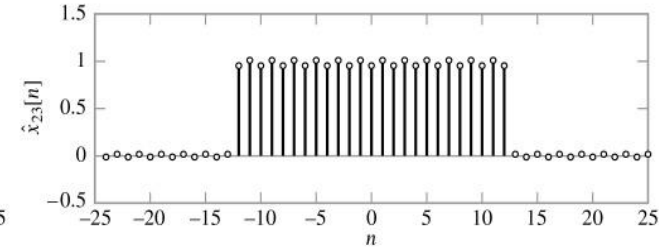
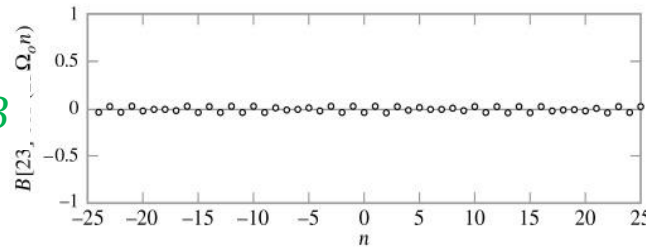
(b)

$J=3$



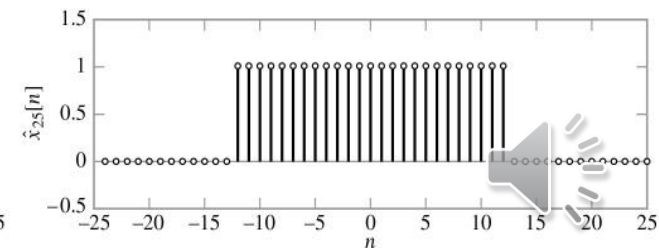
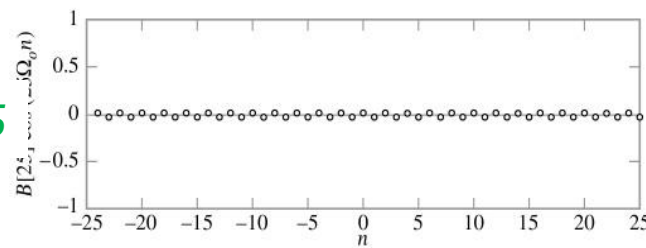
(c)

$J=23$



(d)

$J=25$

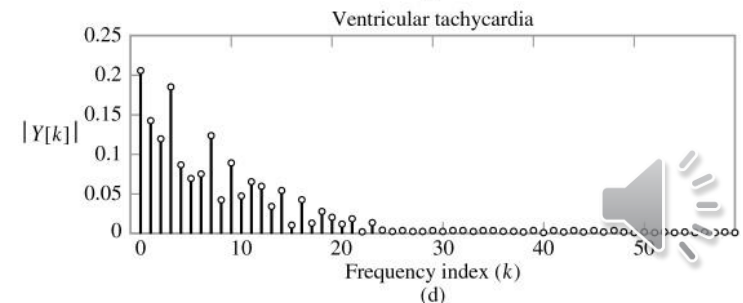
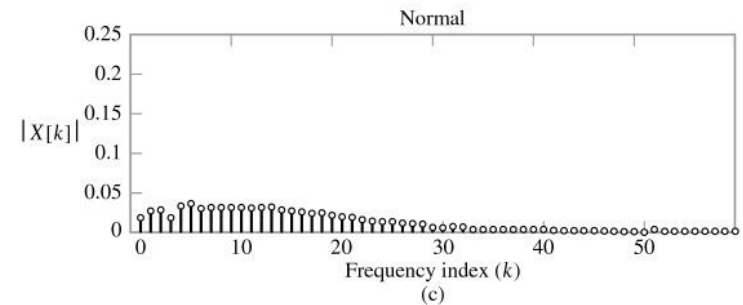
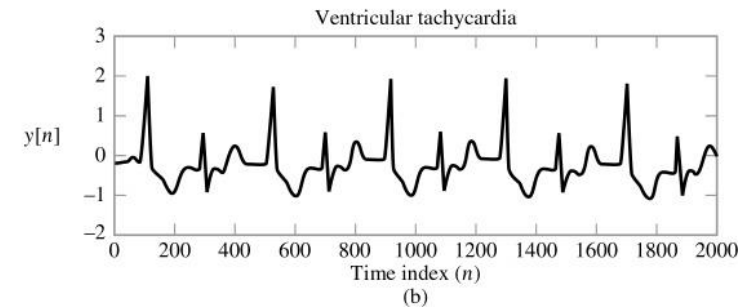
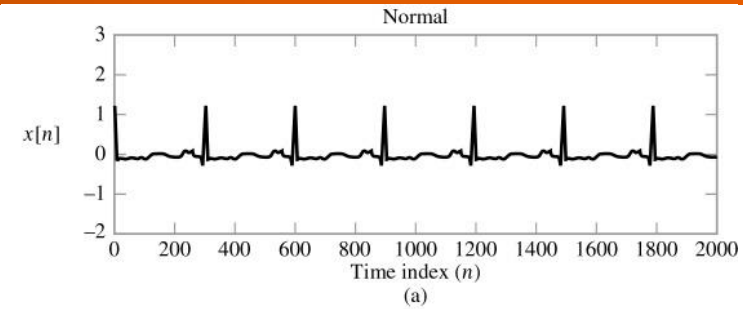


(e)

3.4 D.T. Periodic Signals: D.T. Fourier Series

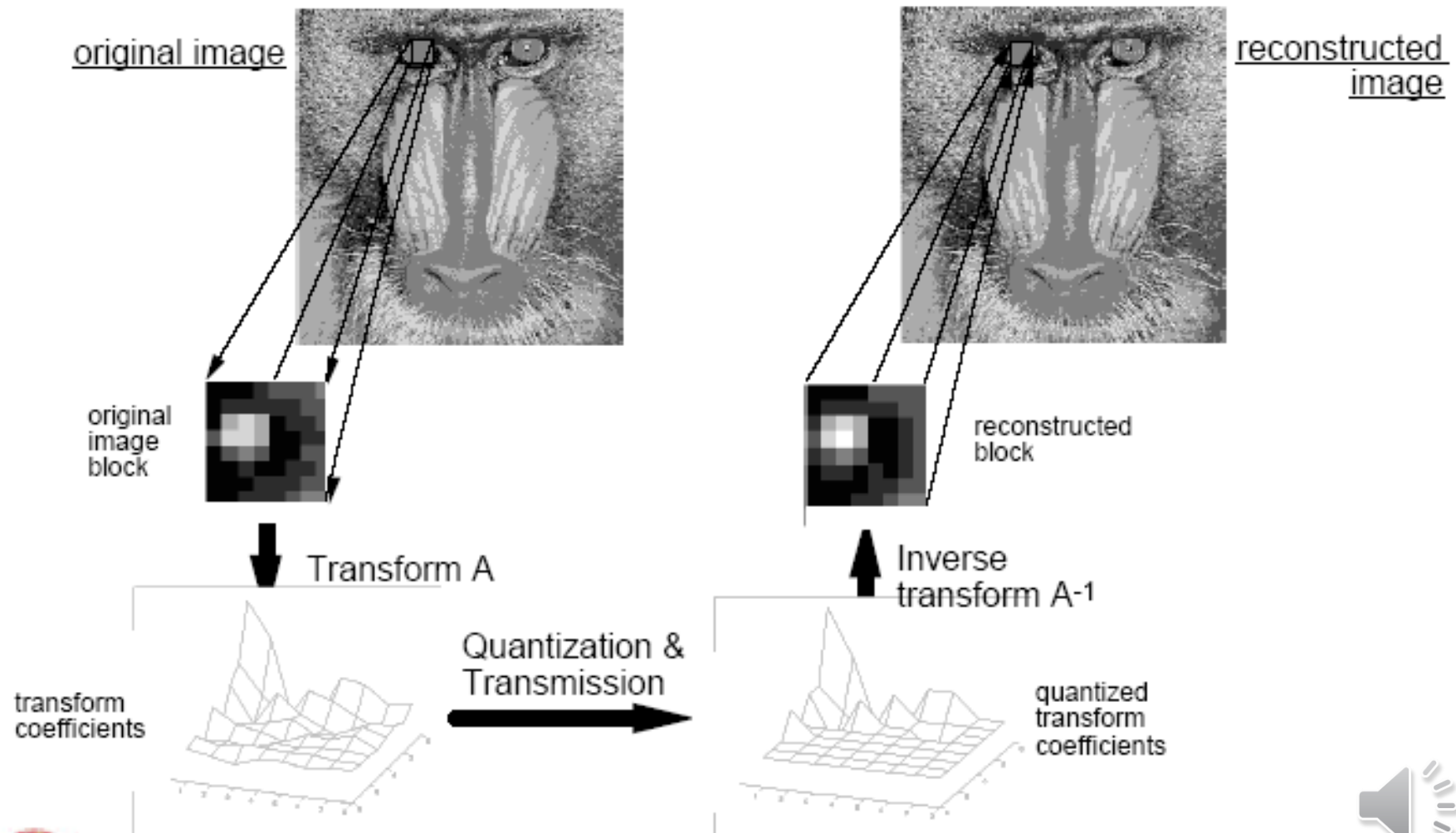
Ex) Electrocardiogram(ECG) waveform

- normal
- ventricular complexes



3.4 D.T. Periodic Signals: D.T. Fourier Series

- Application: Transform coding



3.5 C.T. Periodic Signals: The Fourier Series

- Approximate a signal $x(t)$ having fundamental period T using the series of

$$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_0 t}, \text{ where } \omega_0 = 2\pi / T$$

$$IF \quad \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt < \infty \Rightarrow \frac{1}{T} \int_{\langle T \rangle} |x(t) - \hat{x}(t)|^2 dt = 0$$

- For Pointwise convergence of $x(t)$ and $\hat{x}(t)$, Dirchletet conditions needs except at discontinuities
 - $X(t)$ is bounded
 - $X(t)$ has a finite maxima and minima in one period
 - $X(t)$ has a finite discontinuities in one period



3.6 D.T. Periodic Signals: D.T. Fourier Series

■ Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\omega_0 t}$$

$$X[k] = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt$$

$$x(t) \overset{FS; \omega_0}{\longleftrightarrow} X[k]$$

■ Find FS coefficients: Use orthogonality property

$$\int_{\langle T \rangle} x(t) e^{-jm\omega_0 t} dt = \int_{\langle T \rangle} \left(\sum_{k=-\infty}^{\infty} X[k] e^{jk\omega_0 t} \right) e^{-jm\omega_0 t} dt$$

$$= \sum_{k=-\infty}^{\infty} X[k] \int_{\langle T \rangle} e^{jk\omega_0 t} e^{-jm\omega_0 t} dt$$

$$= X[k]T,$$

$$X[k] = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt$$



3.5 C.T. Periodic Signals: The Fourier Series

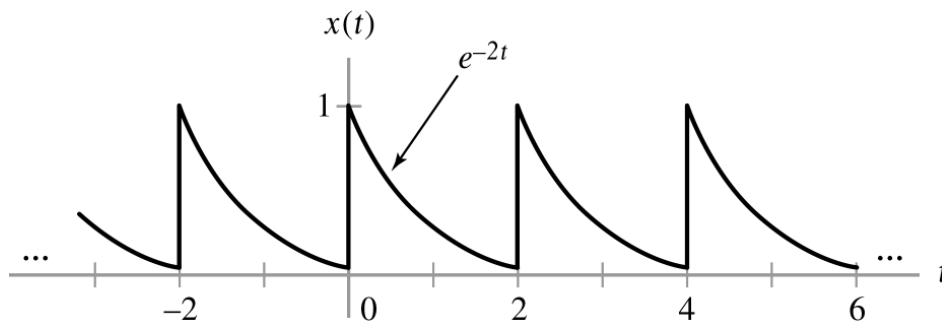
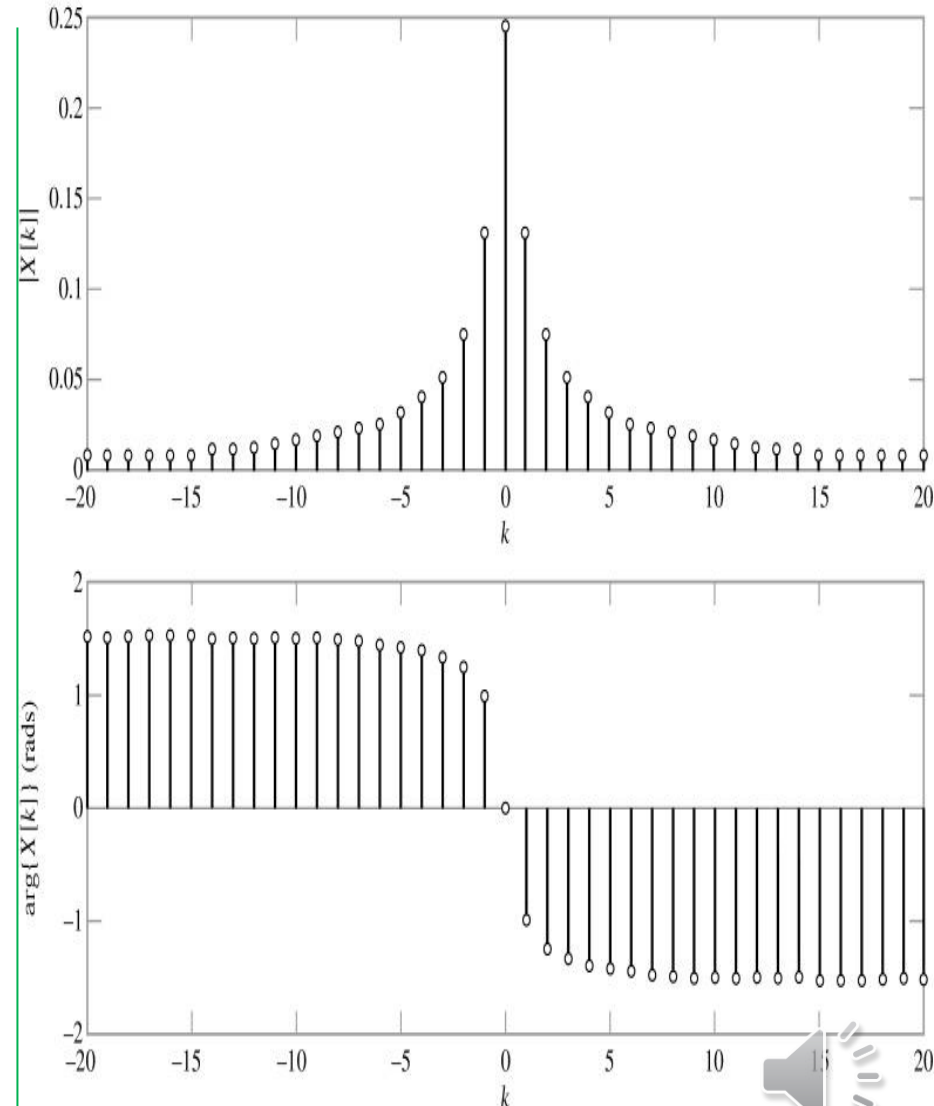


Figure 3.16 (p. 216)

Time-domain signal for Example 3.9.

$$\begin{aligned}
 X[k] &= \frac{1}{2} \int_0^2 e^{-2t} e^{-jk\pi t} dt \\
 &= \frac{1}{2} \int_0^2 e^{-(2+jk\pi)t} dt \\
 &= \frac{-1}{2(2+jk\pi)} e^{-(2+jk\pi)t} \bigg|_0^2 \\
 &= \frac{1-e^{-4}}{4+j2\pi k}, \quad \because e^{-j2k\pi} = 1
 \end{aligned}$$

Ex) FS coeff. by direct calculation



Magnitude and phase spectra for Example 3.9.

3.5 C.T. Periodic Signals: The Fourier Series

Ex) FS coeff. by inspection

$$x(t) = 3 \cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$$

sol)

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\pi t/2} \Rightarrow x(t) = 3 \cdot \frac{1}{2} \left(e^{-j\pi/4} e^{-j(\pi/2)t} + e^{j\pi/4} e^{j(\pi/2)t} \right)$$

$$X[k] = \begin{cases} (3/2)e^{-j\pi/4}, & k = -1 \\ (3/2)e^{j\pi/4}, & k = 1 \\ 0, & \text{otherwise} \end{cases}$$

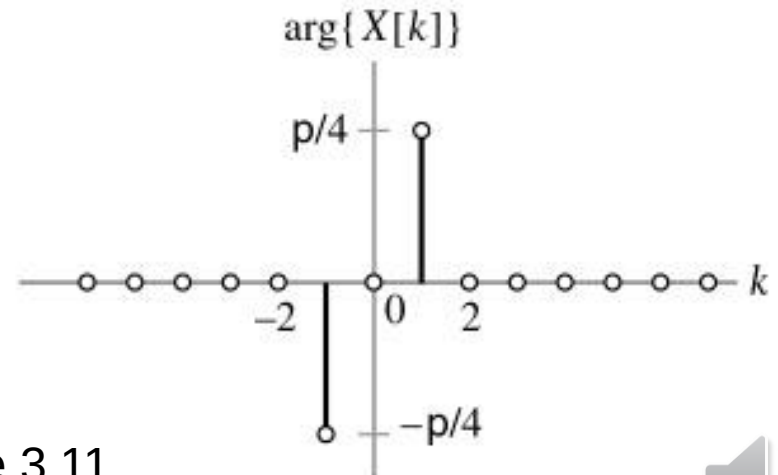
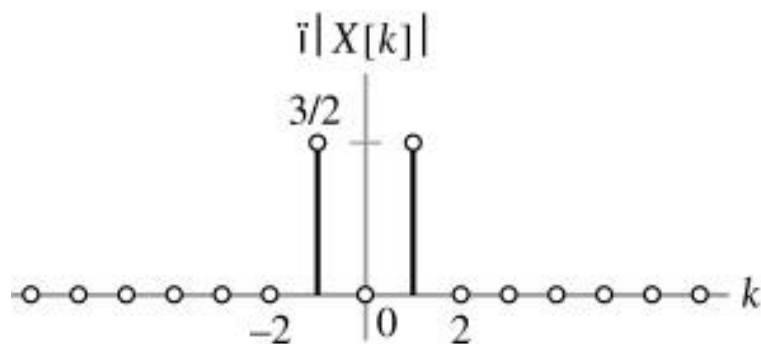


Figure 3.18 (p. 219)

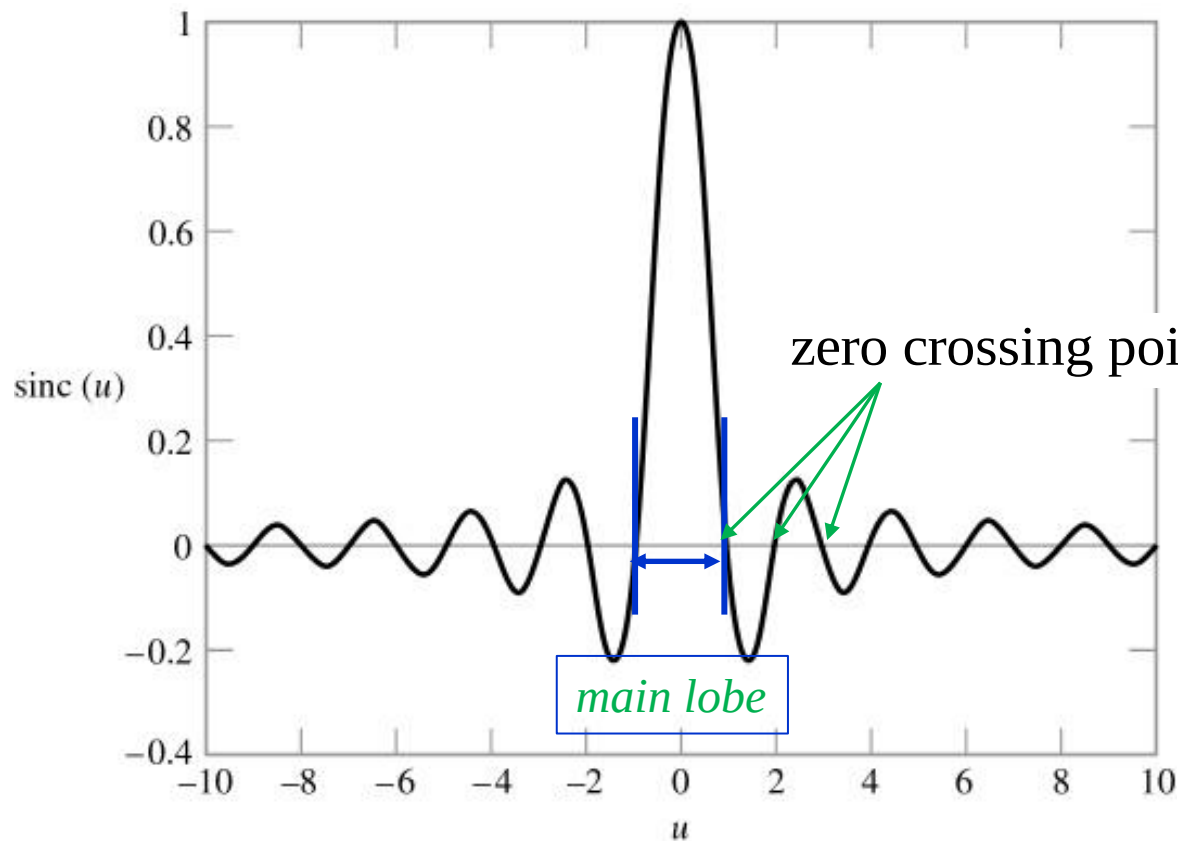
Magnitude and phase spectra for Example 3.11.



3.5 C.T. Periodic Signals: The Fourier Series

cf) Sinc Function

$$\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$$



cf) previous square wave

$$\begin{aligned} & \frac{2}{k\omega_0 T} \sin(k\omega_0 T_0) \\ &= \frac{1}{k\pi} \sin(k2\pi T_0/T), \quad \because \omega_0 = \frac{2\pi}{T} \\ &= \frac{2T_0}{T} \frac{\sin(2\pi k T_0/T)}{2\pi k T_0/T} \\ &= \frac{2T_0}{T} \text{sinc}(2k T_0/T) \end{aligned}$$

$$X[k] = \frac{2T_0}{T} \text{sinc}\left(k \frac{2T_0}{T}\right)$$



3.5 C.T. Periodic Signals: The Fourier Series

cf) Sinc Function

♦ Fourier series pair — Trigonometric FS

1. Trigonometric FS of real-valued signal $x(t)$:

$$x(t) = B[0] + \sum_{k=1}^{\infty} B[k] \cos(k\omega_0 t) + A[k] \sin(k\omega_0 t)$$

FS coefficients:

$$B[0] = \frac{1}{T} \int_0^T x(t) dt$$

$$B[k] = \frac{2}{T} \int_0^T x(t) \cos(k\omega_0 t) dt$$

$$A[k] = \frac{2}{T} \int_0^T x(t) \sin(k\omega_0 t) dt$$

$B[0] = X[0]$ represents the time-averaged value of the signal.

2. Relation between exponential FS and trigonometric FS coefficients:

for $k \neq 0$ (3.27)

Euler's Formula

$$B[k] = X[k] + X[-k]$$

$$A[k] = j(X[k] - X[-k])$$



3.5 C.T. Periodic Signals: The Fourier Series

- If $X[k]$ is symmetric

$$x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{jk\omega_0 t} \quad \because \text{square wave is an even function}$$

$$= X[0] + \sum_{m=1}^{\infty} (X[m]e^{jm\omega_0 t} + X[m]e^{-jm\omega_0 t})$$

$$= X[0] + \sum_{m=1}^{\infty} 2X[m] \left(\frac{e^{jm\omega_0 t} + e^{-jm\omega_0 t}}{2} \right)$$

$$= X[0] + \sum_{m=1}^{\infty} 2X[m] \cos(m\Omega_0 t)$$

$$\text{Let } B[k] = \begin{cases} X[0], & k = 0 \\ 2X[k], & k \neq 0 \end{cases}$$

Trigonometric FS expression of $x(t)$: $\therefore x(t) = \sum_{k=0}^{\infty} B[k] \cos(k\omega_0 t)$



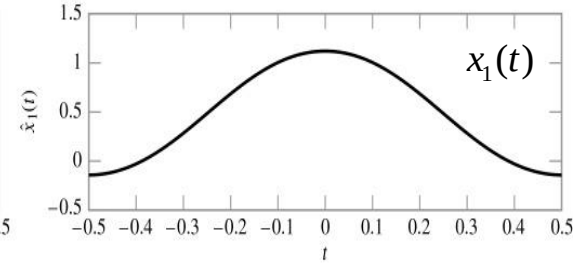
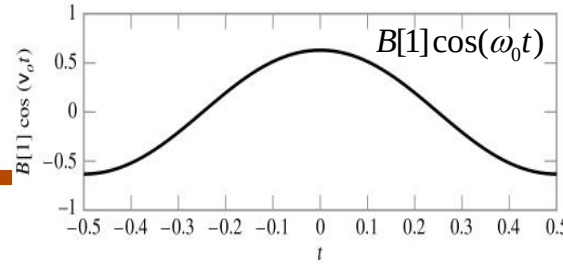
3.5 C.T. Periodic Signals: The Fourier Series

Ex) Partial sum approximation
of square wave

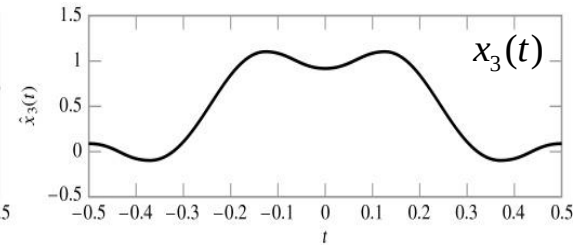
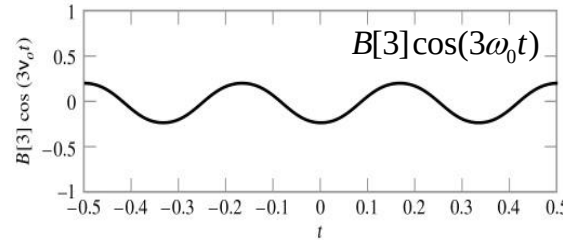
$$\left(\begin{aligned} &\text{Let } T = 1, \quad \frac{T_0}{T} = \frac{1}{4} \\ &\frac{2T_0}{T} \frac{\sin(2\pi k T_0 / T)}{2\pi k T_0 / T} = \frac{1}{2} \frac{\sin(\pi k / 2)}{\pi k / 2} \end{aligned} \right.$$

$$B[k] = \begin{cases} 1/2, & k = 0 \\ 2 \frac{1}{k\pi} (-1)^{(k-1)/2}, & k = \text{odd} \\ 0, & k = \text{even} \end{cases}$$

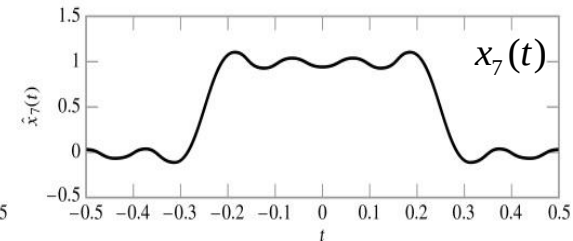
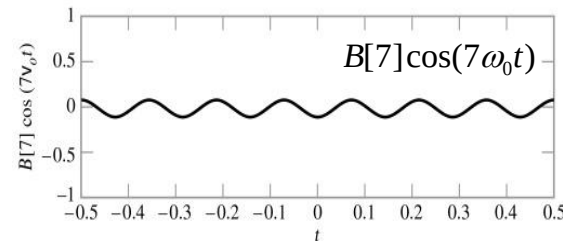
$$x_J(t) = \sum_{k=0}^J B[k] \cos(k\omega_0 t)$$



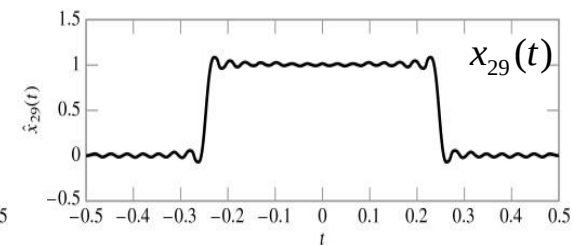
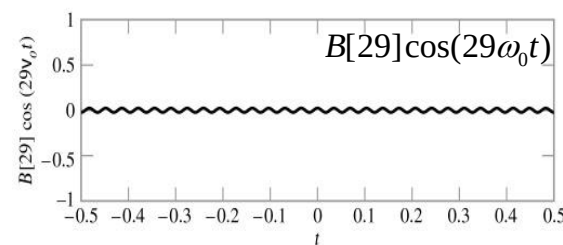
(a)



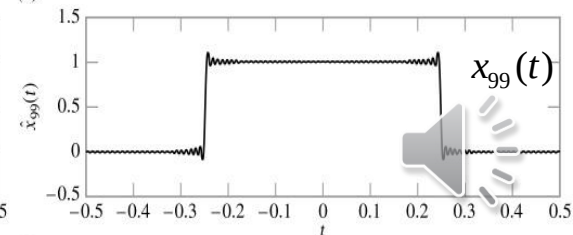
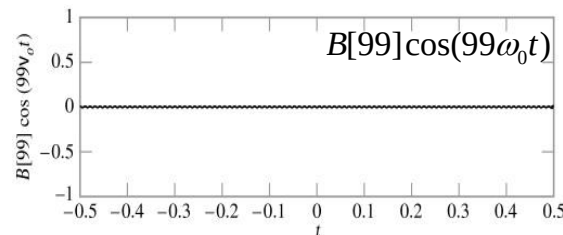
(b)



(c)



(d)



(e)

3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

- Develop the DTFT from the DTFS by a non-periodic signal as the limit of a periodic signal of period N approaches infinity.
- Approximate $x[n]$ with periodic signal.

$$\bullet \quad x[n] = \begin{cases} \tilde{x}[n], & -M \leq n \leq M \\ 0, & |n| > M \end{cases} \Rightarrow x[n] = \lim_{M \rightarrow \infty} \tilde{x}[n], \quad \tilde{x}[n]: \text{Periodic}$$

$$\bullet \quad \tilde{x}[n] = \sum_{k=-M}^M X[k] e^{jk\Omega_0 n} \Leftrightarrow X[k] = \frac{1}{2M+1} \sum_{n=-M}^M \tilde{x}[n] e^{-jk\Omega_0 n}$$

$$\bullet \quad \text{Continuous function of frequency } X(e^{j\Omega})$$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\Omega})$$

$$X(e^{j\Omega}) = \sum_{n=-M}^M x[n] e^{-j\Omega n}$$

$$X[k] = X(e^{jk\Omega_0}) / (2M+1)$$

$$\tilde{x}[n] = \frac{1}{2M+1} \sum_{k=-M}^M X(e^{jk\Omega_0}) e^{jk\Omega_0 n}$$

$$\bullet \quad \text{Using the relation } \Omega_0 = 2\pi / (2M+1)$$

$$\tilde{x}[n] = \frac{1}{2\pi} \sum_{k=-M}^M X(e^{jk\Omega_0}) e^{jk\Omega_0 n} \Omega_0$$

As $M \rightarrow \infty$

$$x[n] = \lim_{M \rightarrow \infty} \frac{1}{2\pi} \sum_{k=-M}^M X(e^{jk\Omega_0}) e^{jk\Omega_0 n} \Omega_0$$

$$\therefore x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$



3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

▪ DTFT:

$$\begin{aligned} \bullet \quad x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega \\ \bullet \quad X(e^{j\Omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \end{aligned}$$

$$x[n] \xleftrightarrow{\text{DTFT}} X[e^{j\Omega}]$$

- If $x[n]$ is absolutely summable, that is,

$$\sum_{n=-\infty}^{n=\infty} |x[n]| < \infty$$

→ the sum of DTFT converges uniformly a *continuous function of Ω* .

- If $x[n]$ is not absolutely summable, but does have finite energy,

$$\sum_{n=-\infty}^{n=\infty} |x[n]|^2 < \infty$$

→ The sum of DTFT converges in a MSE (mean-square error) sense **but does not converge pointwise.**



3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Ex) $x[n] = \alpha^n u[n]$, non-periodic.

$X(e^{j\Omega})$?

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\Omega n}$$

This sum diverges for $|\alpha| \geq 1$

$$= \sum_{n=0}^{\infty} \alpha^n e^{-j\Omega n}$$

$$= \sum_{n=0}^{\infty} (\alpha e^{-j\Omega})^n$$

For $\alpha < 1$,
we have $= \frac{1}{1 - \alpha e^{-j\Omega}}, \quad |\alpha| < 1$

If α is
real valued $\rightarrow X(e^{j\Omega}) = \frac{1}{1 - \alpha \cos \Omega + j\alpha \sin \Omega}$

Even
function

$$|X(e^{j\Omega})| = \frac{1}{((1 - \alpha \cos \Omega)^2 + \alpha^2 \sin^2 \Omega)^{1/2}}$$

Euler's Formula

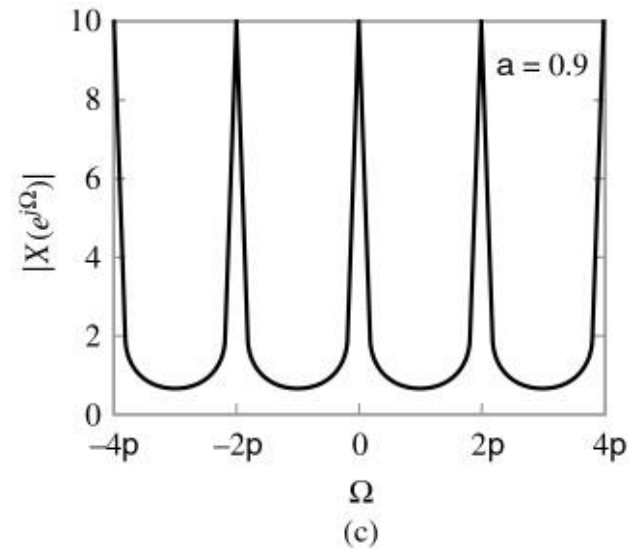
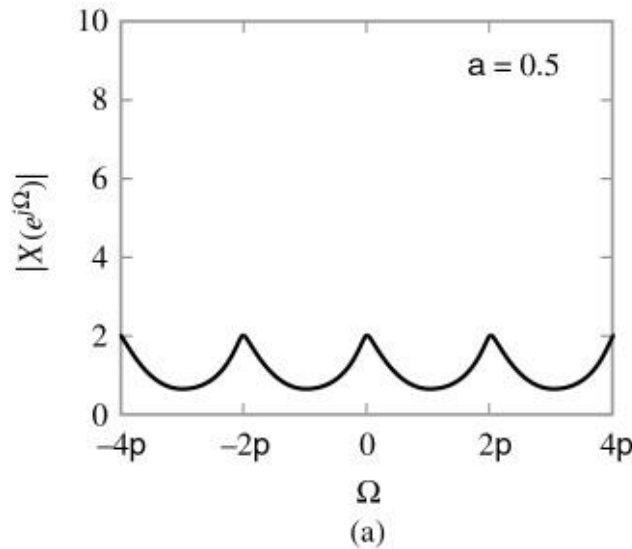
$$\arg\{X(e^{j\Omega})\} = -\arctan\left(\frac{\alpha \sin \Omega}{1 - \alpha \cos \Omega}\right)$$

Odd function

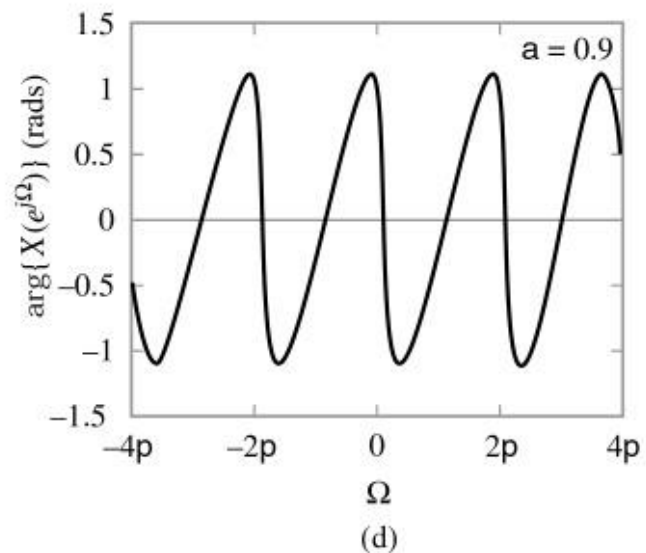
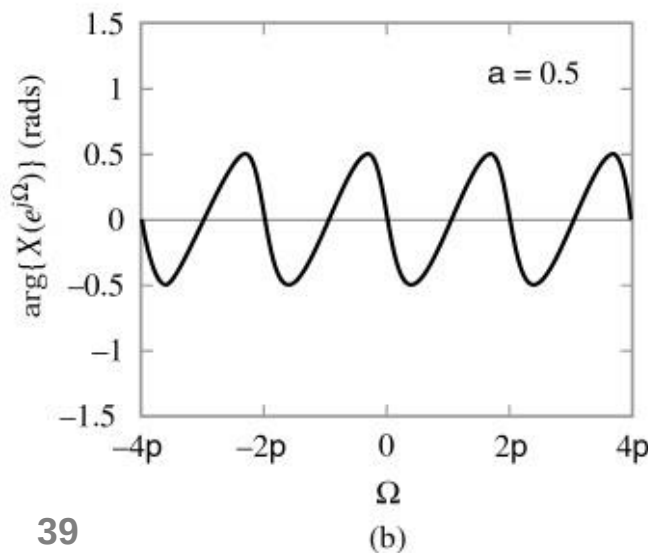
3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Ex) The DTFT of $x[n] = \alpha^n u[n]$

$\alpha = 0.5$



$\alpha = 0.9$



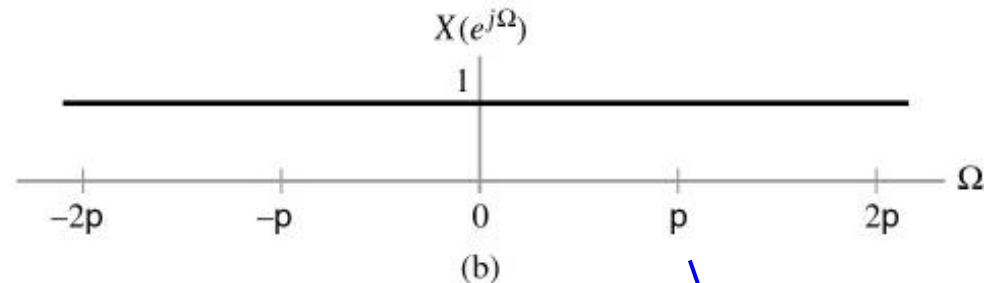
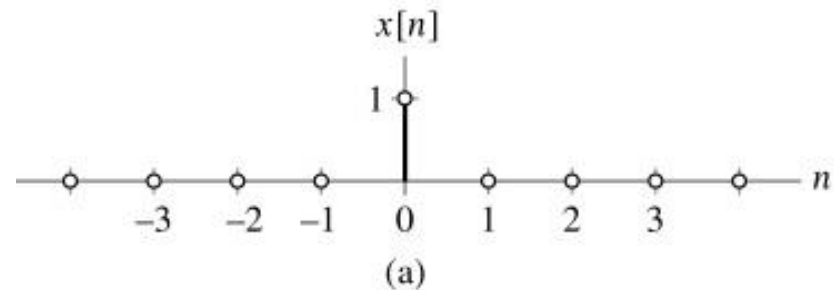
3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Ex) impulse $x[n] = \delta[n]$

(sol)

DTFT of $x[n]$: $X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\Omega n} = 1$

$\delta[n] \xleftrightarrow{\text{DTFT}} 1$



$p \equiv \pi$



3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Ex) Find the inverse DTFT of

$$X(e^{j\Omega}) = \delta(\Omega), -\pi < \Omega \leq \pi$$

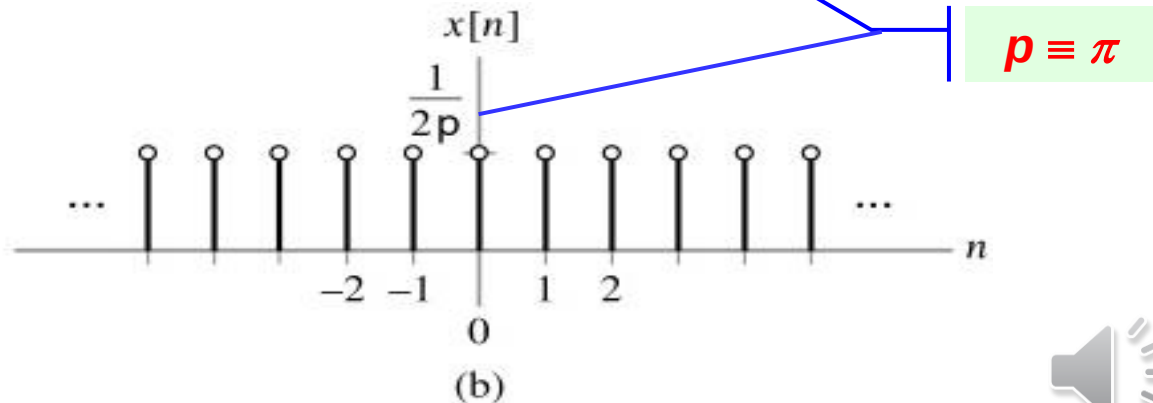
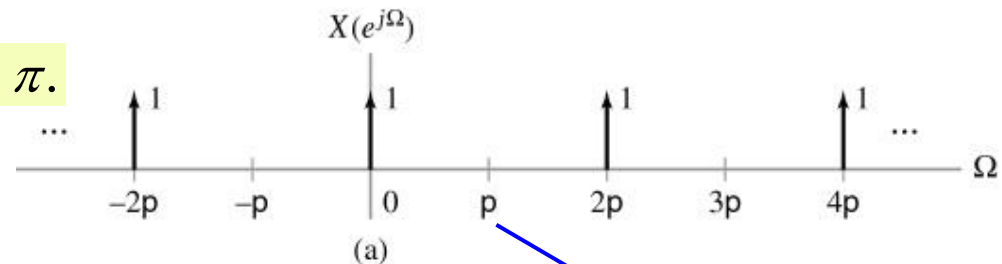
(sol)

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi}$$

Sifting property of impulse function

$$\frac{1}{2\pi} \xleftrightarrow{\text{DTFT}} \delta(\Omega), \quad -\pi < \Omega \leq \pi.$$

$$X(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} \delta(\Omega - k2\pi).$$



$$p \equiv \pi$$



3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Ex) Moving-average system

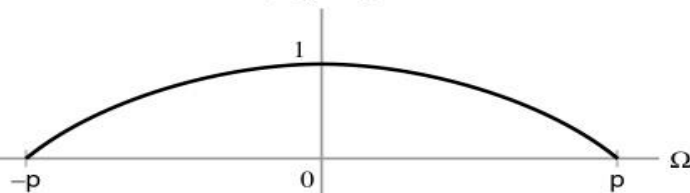
$$y_1[n] = \frac{1}{2}(x[n] + x[n-1])$$

$$h_1[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1]$$

$$H_1(e^{j\Omega}) = \frac{1}{2} + \frac{1}{2}e^{-j\Omega}$$

$$H_1(e^{j\Omega}) = e^{-j\frac{\Omega}{2}} \frac{e^{j\frac{\Omega}{2}} + e^{-j\frac{\Omega}{2}}}{2} = e^{-j\Omega/2} \cos\left(\frac{\Omega}{2}\right)$$

$|H_1(e^{j\Omega})|$



$$p \equiv \pi$$

Magnitude response: Phase response:

$$|H_1(e^{j\Omega})| = \left| \cos\left(\frac{\Omega}{2}\right) \right|, \quad \arg\{H_1(e^{j\Omega})\} = -\frac{\Omega}{2}.$$

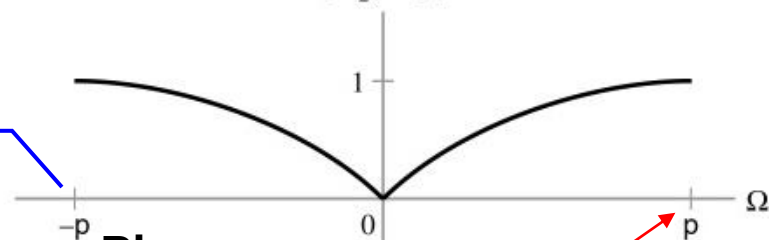
$$y_2[n] = \frac{1}{2}(x[n] - x[n-1])$$

$$h_2[n] = \frac{1}{2}\delta[n] - \frac{1}{2}\delta[n-1]$$

$$H_2(e^{j\Omega}) = \frac{1}{2} - \frac{1}{2}e^{-j\Omega}$$

$$= je^{-j\Omega/2} \sin\left(\frac{\Omega}{2}\right)$$

$|H_2(e^{j\Omega})|$



Phase response:

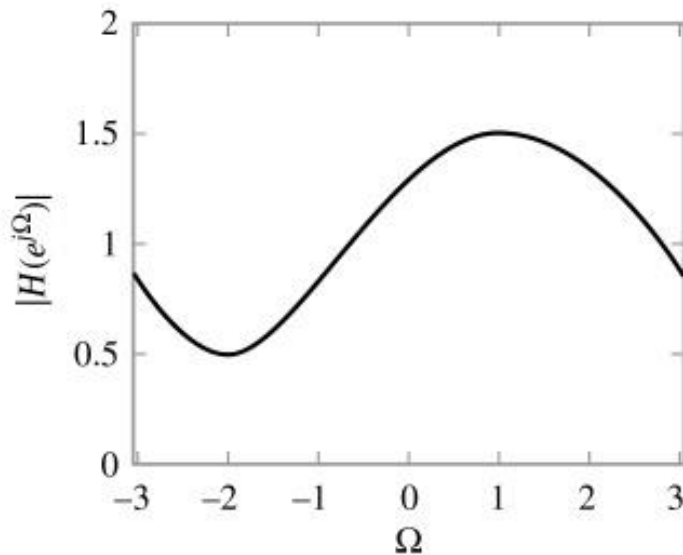
$$\arg\{H_2(e^{j\Omega})\} = \begin{cases} \pi/2 - \frac{\Omega}{2}, & \Omega > 0 \\ -\pi/2 - \frac{\Omega}{2}, & \Omega < 0 \end{cases}$$

3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

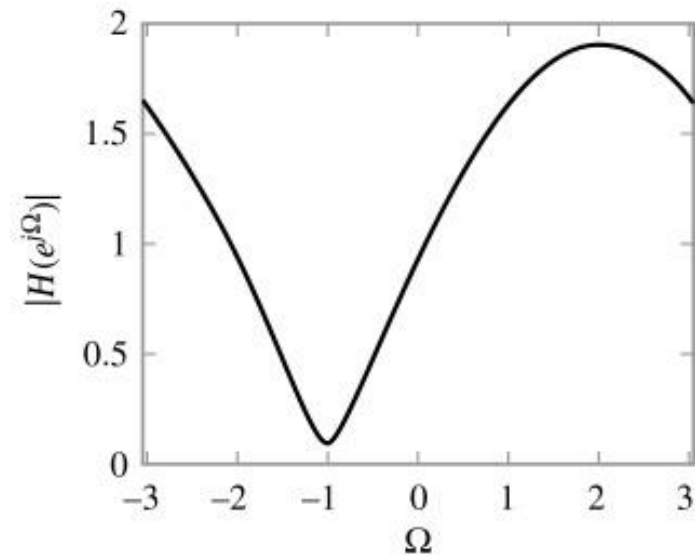
- Multipath communication channel

$$y[n] = x[n] + ax[n-1]$$

$$H(e^{j\Omega}) = 1 + ae^{-j\Omega}$$



(a) Echo coefficient $a = 0.5e^{j\pi/3}$



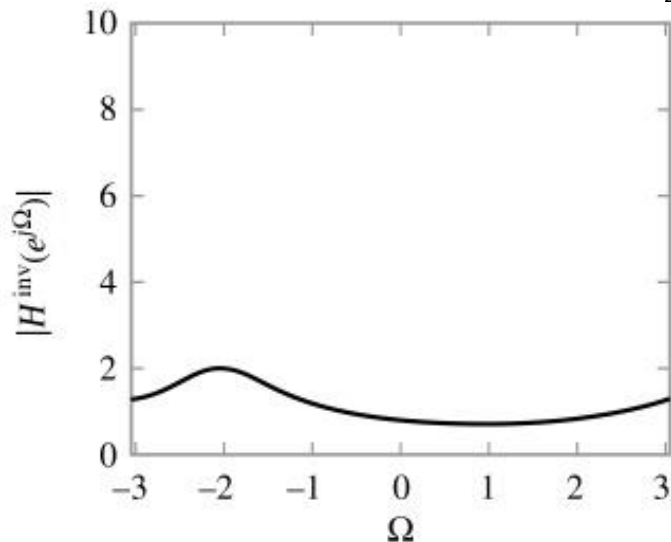
(b) Echo coefficient $a = 0.9e^{j2\pi/3}$.



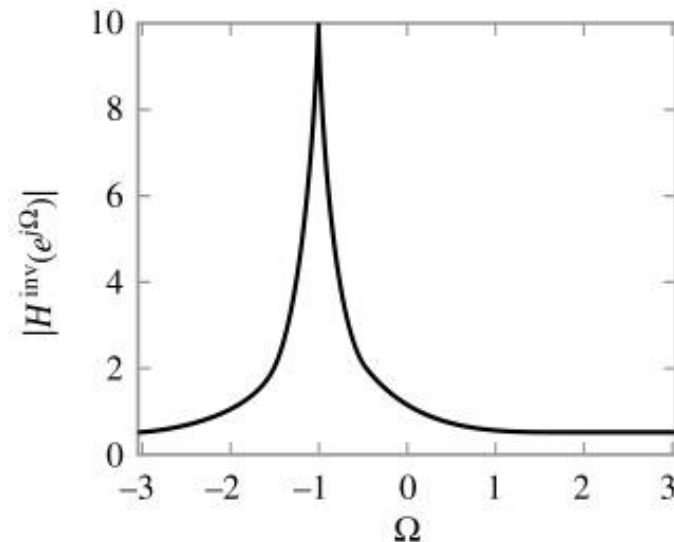
3.6 D.T. Nonperiodic Signals : D.T. Fourier Transform

Figure 3.38 (p. 241)

Magnitude response of the inverse system for multipath propagation in Example 3.23. (a) Echo coefficient $a = 0.5e^{j\pi/3}$. (b) Echo coefficient $a = 0.9e^{j\pi/3}$



(a)



(b)

$$\left| H^{\text{inv}}(e^{j\Omega}) \right| = \frac{1}{\left(1 + |a|^2 + 2|a| \cos(\Omega - \arg\{a\}) \right)^{1/2}}.$$



3.7 C.T. Nonperiodic Singals: Fourier Transform

- The FT is expressed as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Notation for FT pair:

$$x(t) \overset{FT}{\longleftrightarrow} X(j\omega)$$

Frequency-domain representation of the signal $x(t)$

- Convergence

- Square integrable

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

: MSE between $x(t)$ and $\bar{x}'(t)$, where Approximation:

$$x'(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

← Zero MSE does not imply pointwise convergence



3.7 C.T. Nonperiodic Singals: Fourier Transform

- Dirichlet condition

- Absolute integrable

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

- A finite number of local maxima, minima, and discontinuities in any finite interval

- The size of each discontinuity is finite

→ Pointwise convergence at all values of t except those corresponding to discontinuities.



3.7 C.T. Nonperiodic Singals: Fourier Transform

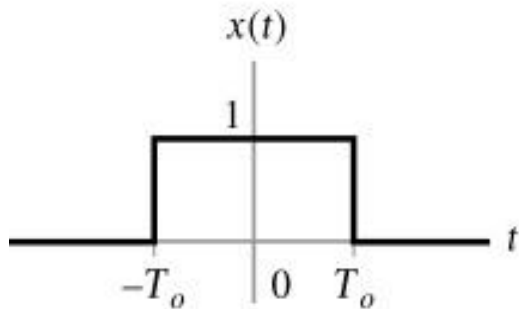
Ex)

$$x(t) = \begin{cases} 1, & -T_0 \leq t \leq T_0 \\ 0, & |t| > T_0 \end{cases}$$

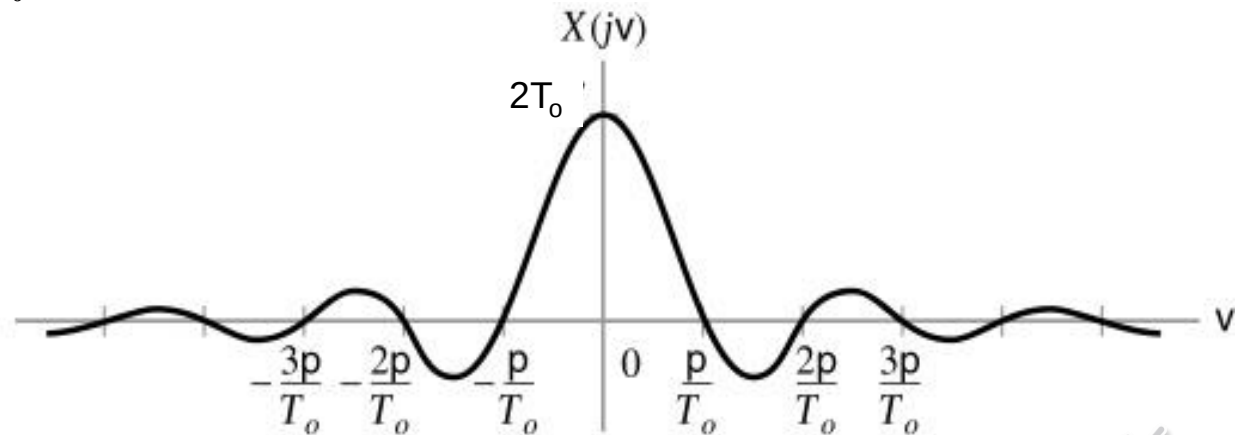
(sol)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_{-T_0}^{T_0} e^{-j\omega t} dt$$

$$= -\frac{1}{j\omega} e^{-j\omega t} \Big|_{-T_0}^{T_0} = \frac{2}{\omega} \sin(\omega T_0) = 2T_0 \frac{\sin(\omega T_0)}{\omega T_0} = 2T_0 \text{sinc}\left(\frac{2T_0}{T}\right)$$



(a)



(b)



■ Summary of F.T., F.S. for C.T. D.T. with periodic and nonperiodic

<i>Time.</i>	<i>Periodic</i>	<i>Nonperiodic</i>	
<i>Continuous</i> <i>(t)</i>	<i>Fourier Series</i> $x(t) = \sum_{k=-\infty}^{\infty} X[k]e^{j\omega_0 t}$ $X[k] = \frac{1}{T} \int_{\langle T \rangle} x(t)e^{-jk\omega_0 t} dt$ $x(t) \text{ has period } T, \omega_0 = \frac{2\pi}{T}$	<i>Fourier Transform</i> $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$ $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$	<i>Nonperiodic</i> <i>(k, \omega)</i>
<i>Discrete</i> <i>[n]</i>	<i>Discrete-Time Fourier Series</i> $x[n] = \sum_{k=\langle N \rangle} X[k]e^{jk\Omega_0 n}$ $X[k] = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]e^{-jk\Omega_0 n}$ $x[n] \text{ and } X[k] \text{ have period } N$ $\Omega_0 = \frac{2\pi}{N}$	<i>Discrete-Time Fourier Transform</i> $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n} d\Omega$ $X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$ $X(e^{j\Omega}) \text{ has period } 2\pi$	<i>Periodic</i> <i>(k, \Omega)</i>
	<i>Discrete</i> <i>[k]</i>	<i>Continuous</i> <i>(\omega, \Omega)</i>	



3.8 Properties of Fourier Representation

- Periodicity Properties

Time-Domain Properties		Frequency-Domain Properties	
Continuous	↔	Nonperiodic	
Discrete	↔	Periodic	
Periodic	↔	Discrete	
Nonperiodic	↔	Continuous	



3.9 Linearity and Symmetric Properties

- **Linearity:** All Fourier representations are linear operator

$$z(t) = ax(t) + by(t) \xleftrightarrow{FT} Z(j\omega) = aX(j\omega) + bY(j\omega)$$

$$z(t) = ax(t) + by(t) \xleftrightarrow{FS; \omega_0 = \frac{2\pi}{T}} Z(j\omega) = aX[k] + bY[k]$$

$$z[n] = ax[n] + by[n] \xleftrightarrow{DTFT} Z(e^{j\Omega}) = aX(e^{j\Omega}) + bY(e^{j\Omega})$$

$$z[n] = ax[n] + by[n] \xleftrightarrow{DTFS; \Omega_0 = \frac{2\pi}{N}} Z(j\omega) = aX[k] + bY[k]$$



3.9 Linearity and Symmetric Properties

■ Symmetry Properties

- $x(t)$ is real,

$$\begin{aligned} X^*(j\omega) &= \left[\int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt \right]^* \\ &= \int_{-\infty}^{\infty} x^*(t)e^{j\omega t} dt = \int_{-\infty}^{\infty} x(t)e^{-j(-\omega)t} dt \\ &= X(-j\omega) \end{aligned}$$

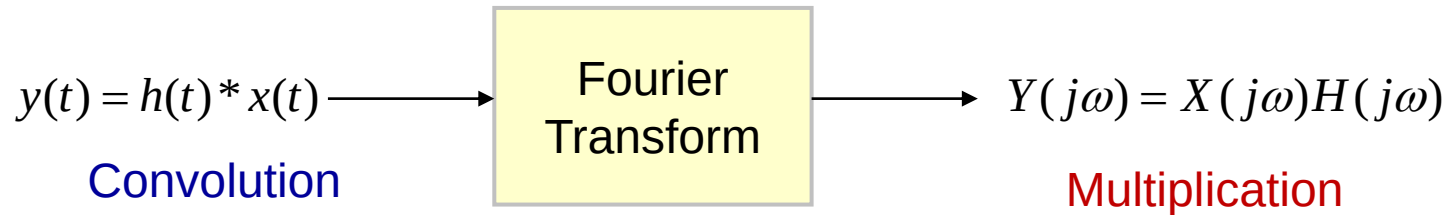
- $x(t)$ is imaginary

$$\begin{aligned} X^*(j\omega) &= \left[\int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt \right]^* \\ &= \int_{-\infty}^{\infty} x^*(t)e^{j\omega t} dt = - \int_{-\infty}^{\infty} x(t)e^{-j(-\omega)t} dt \\ &= -X(-j\omega) \end{aligned}$$

Representation	Real-valued time signal	Imaginary valued time signal
FT	$X^*(j\omega) = X(-j\omega)$	$X^*(j\omega) = -X(-j\omega)$
FS	$X^*[k] = X[-k]$	$X^*[k] = -X[-k]$
DTFT	$X^*(e^{j\Omega}) = X(e^{-j\Omega})$	$X^*(e^{j\Omega}) = -X(e^{-j\Omega})$
DTFS	$X^*[k] = X[-k]$	$X^*[k] = -X[-k]$



3.10 Convolution Property



$$\begin{aligned}
 y(t) &= h(t) * x(t) \\
 &= \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \\
 x(t - \tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega(t - \tau)} d\omega
 \end{aligned}$$

$$\begin{aligned}
 y(t) &= \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \\
 &= \int_{-\infty}^{\infty} h(\tau) \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega(t - \tau)} d\omega d\tau \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau X(j\omega) e^{j\omega t} d\omega \\
 \therefore y(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\omega) X(j\omega) e^{j\omega t} d\omega
 \end{aligned}$$

$$y(t) = h(t) * x(t) \overset{\text{CTFT}}{\leftrightarrow} Y(j\omega) = X(j\omega)H(j\omega)$$

$$y[n] = x[n] * h[n] \overset{\text{DTFT}}{\leftrightarrow} Y(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})$$



3.10 Convolution Property

A convolution in the frequency domain

$$\text{Ex) } y(t) = x(t) * h(t)$$

$$x(t) = (1/(\pi t)) \sin(\pi t)$$

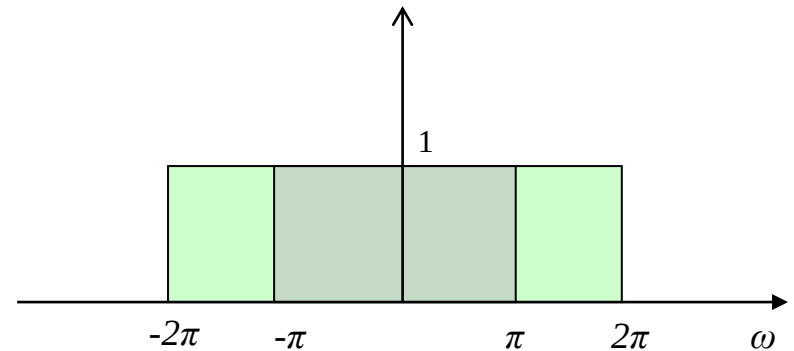
$$h(t) = (1/(\pi t)) \sin(2\pi t)$$

$$x(t) \xleftrightarrow{FT} X(j\omega) = \begin{cases} 1, & |\omega| < \pi \\ 0, & |\omega| > \pi \end{cases}$$

$$h(t) \xleftrightarrow{FT} H(j\omega) = \begin{cases} 1, & |\omega| < 2\pi \\ 0, & |\omega| > 2\pi \end{cases}$$

$$x(t) * h(t) \xleftrightarrow{FT} X(j\omega)H(j\omega)$$

$$Y(j\omega) = \begin{cases} 1, & |\omega| < \pi \\ 0, & |\omega| > \pi \end{cases} \rightarrow y(t) = (1/(\pi t)) \sin(\pi t)$$



3.10 Convolution Property

Filtering

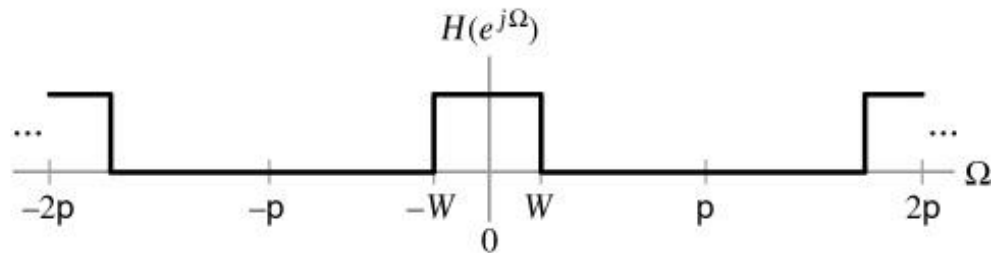
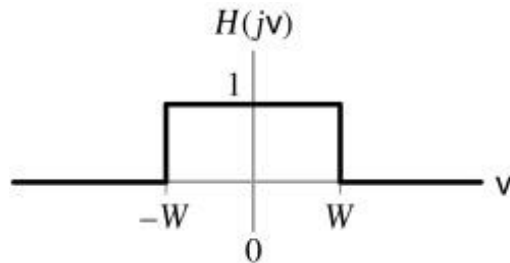
- Filtering : multiplication that occurs in the frequency-domain representation
- A system performs filtering on the signal by presenting a different response to components of the input that are at different frequencies.
- The term, filtering, implies that some frequency components of the input are eliminated while others are passed by the system unchanged.



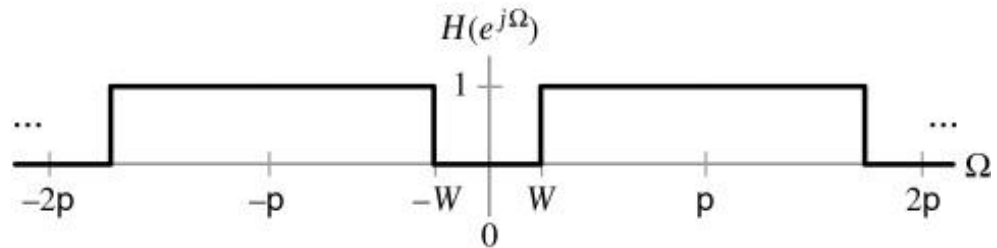
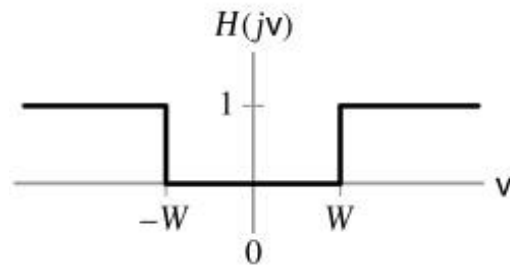
3.10 Convolution Property

Frequency response of ideal CT filters

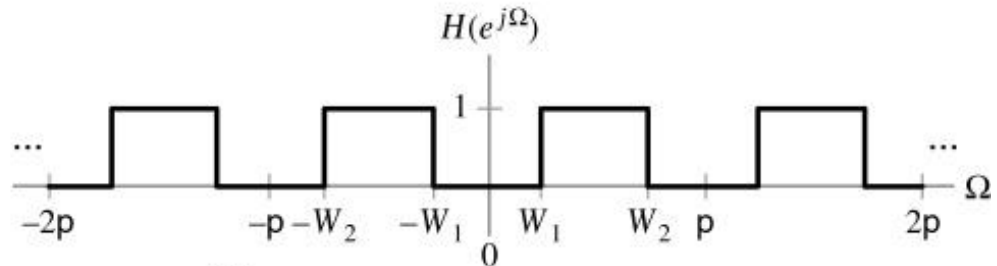
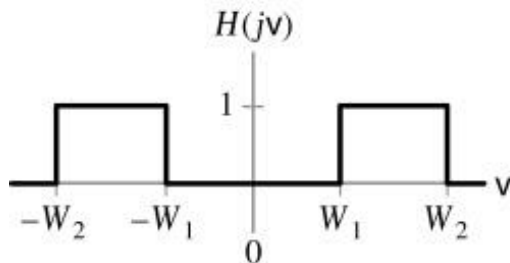
Frequency response of ideal DT filters



(a) Low-pass characteristic



(b) High-pass characteristic



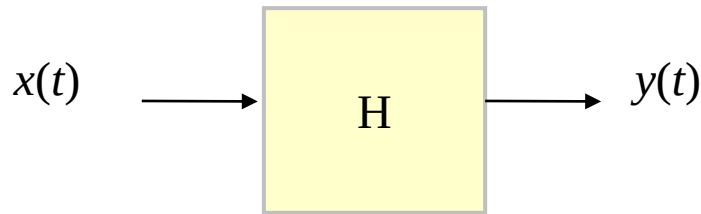
(c) Band-pass characteristic



3.10 Convolution Property

▪ System Identification

If the spectrum is nonzero at all frequencies, the frequency response of a system can be determined from the knowledge of the input and output spectra.



$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$$

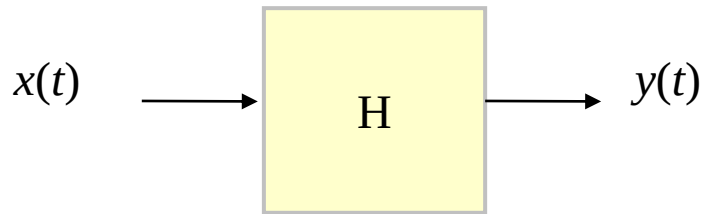
Ex) $x(t) = e^{-2t}u(t)$
 $y(t) = e^{-t}u(t)$
 $h(t) = ?$



3.10 Convolution Property

▪ System Identification

If the spectrum is nonzero at all frequencies, the frequency response of a system can be determined from the knowledge of the input and output spectra.



$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$$

Ex) $x(t) = e^{-2t}u(t)$

$$y(t) = e^{-t}u(t)$$

$$h(t) = ?$$

Sol) $X(j\omega) = \frac{1}{j\omega + 2}, \quad Y(j\omega) = \frac{1}{j\omega + 1}$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = 1 + \frac{1}{j\omega + 1},$$

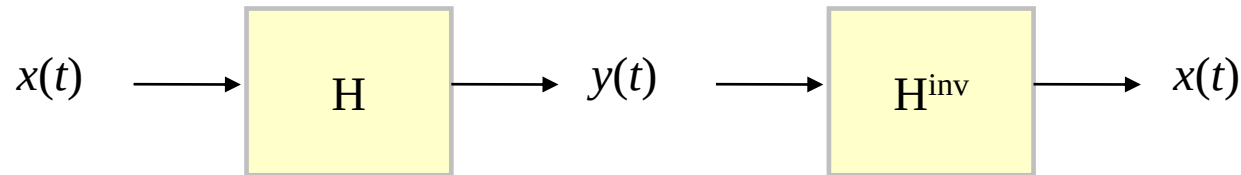
$$h(t) = \delta(t) + e^{-t}u(t)$$



3.10 Convolution Property

- Inverse System**

Recover the input of the system from the output



$$X(j\omega) = H^{inv}Y(j\omega) = \frac{1}{H(j\omega)}Y(j\omega)$$

Ex) equalization

$$y[n] = x[n] + ax[n-1], |a| < 1$$

Sol)

$$h[n] = \begin{cases} 1, n = 0 \\ a, n = 1 \\ 0, \text{ohterwise} \end{cases}$$
$$H(e^{j\Omega}) = 1 + ae^{-j\Omega}$$
$$H^{inv}(e^{j\Omega}) = \frac{1}{1 + ae^{-j\Omega}}$$

$$\therefore h^{inv}[n] = (-a)^n u[n]$$



3.10 Convolution Property

- Periodic convolution and (DT)FS representations

$$C, NP: \quad x(t) * z(t) \xleftrightarrow{FT} X(j\omega)Z(j\omega)$$

$$C, P: \quad x(t) \boxtimes z(t) \xleftrightarrow{FS; \omega_0} T X[k]Z[k]$$

$$D, NP: \quad x[n] * z[n] \xleftrightarrow{DTFT} X(e^{j\Omega})Z(e^{j\Omega})$$

$$D, P: \quad x[n] \boxtimes z[n] \xleftrightarrow{DTFS; \Omega_0} N X[k]Z[k]$$



3.11 Differentiation and Integration Properties

▪ Differentiation in Time

- Differentiation of (non)periodic signal in time

CT nonperiodic

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) j\omega e^{j\omega t} d\omega$$

$$\frac{d}{dt} x(t) \xleftrightarrow{FT} j\omega X(j\omega)$$

CT periodic

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\omega_0 t}$$

$$\frac{d}{dt} x(t) = \sum_{k=-\infty}^{\infty} X[k] jk\omega_0 e^{jk\omega_0 t}$$

$$\frac{d}{dt} x(t) \xleftrightarrow{FS:\omega_0} jk\omega_0 X[k]$$



3.11 Differentiation and Integration Properties

▪ Differentiation in Frequency

- Differentiation of (non)periodic signal in frequency

nonperiodic

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} -jtx(t)e^{-j\omega t} dt$$

$$-jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega} X(j\omega)$$

periodic

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

$$-jnx[n] \xleftrightarrow{FT} \frac{d}{d\Omega} X(e^{j\Omega})$$



3.12 Time- and Frequency-Shift Properties

- Time-shift properties of Fourier representation

$$x(t - t_0) \overset{FT}{\longleftrightarrow} e^{-j\omega t_0} X(j\omega)$$

$$x(t - t_0) \overset{FS; \omega_0}{\longleftrightarrow} e^{-jk\omega_0 t_0} X[k]$$

$$x[n - n_0] \overset{DTFT}{\longleftrightarrow} e^{-j\Omega n_0} X(e^{j\Omega})$$

$$x[n - n_0] \overset{DTFS; \Omega_0}{\longleftrightarrow} e^{-jk\Omega_0 n_0} X[k]$$



3.12 Time- and Frequency-Shift Properties

- Frequency-shift properties of Fourier representation

$$e^{j\gamma t} x(t) \xleftrightarrow{FT} X(j(\omega - \gamma))$$

$$e^{jk_0\omega_0 t} x(t) \xleftrightarrow{FS;\omega_0} X[k - k_0]$$

$$e^{j\Gamma n} x[n] \xleftrightarrow{DTFT} X(e^{j(\Omega - \Gamma)})$$

$$e^{jk_0\Omega_0 n} x[n] \xleftrightarrow{DTFS;\Omega_0} X[k - k_0]$$



3.12 Time- and Frequency-Shift Properties

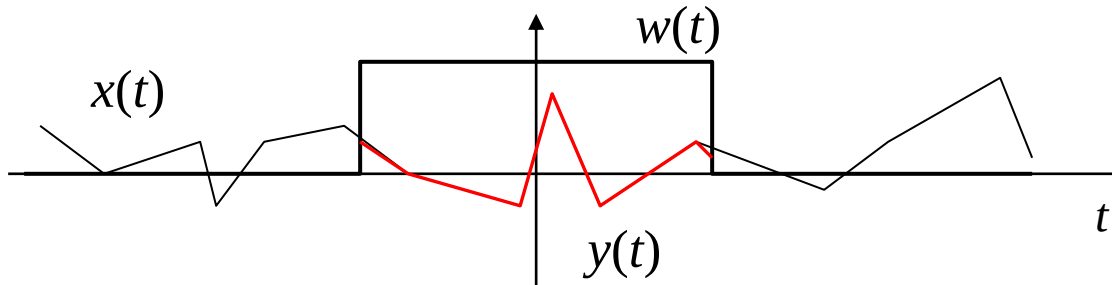
Ex)

$$z(t) = \begin{cases} e^{j10t}, & |t| \leq \pi \\ 0, & \text{otherwise} \end{cases}$$



3.14 Multiplication Property

Windowing in time domain → Truncating



$$y(t) = x(t)w(t)$$

$$y(t) \xleftrightarrow{FT} Y(j\omega) = \frac{1}{2\pi} X(j\omega) * W(j\omega)$$

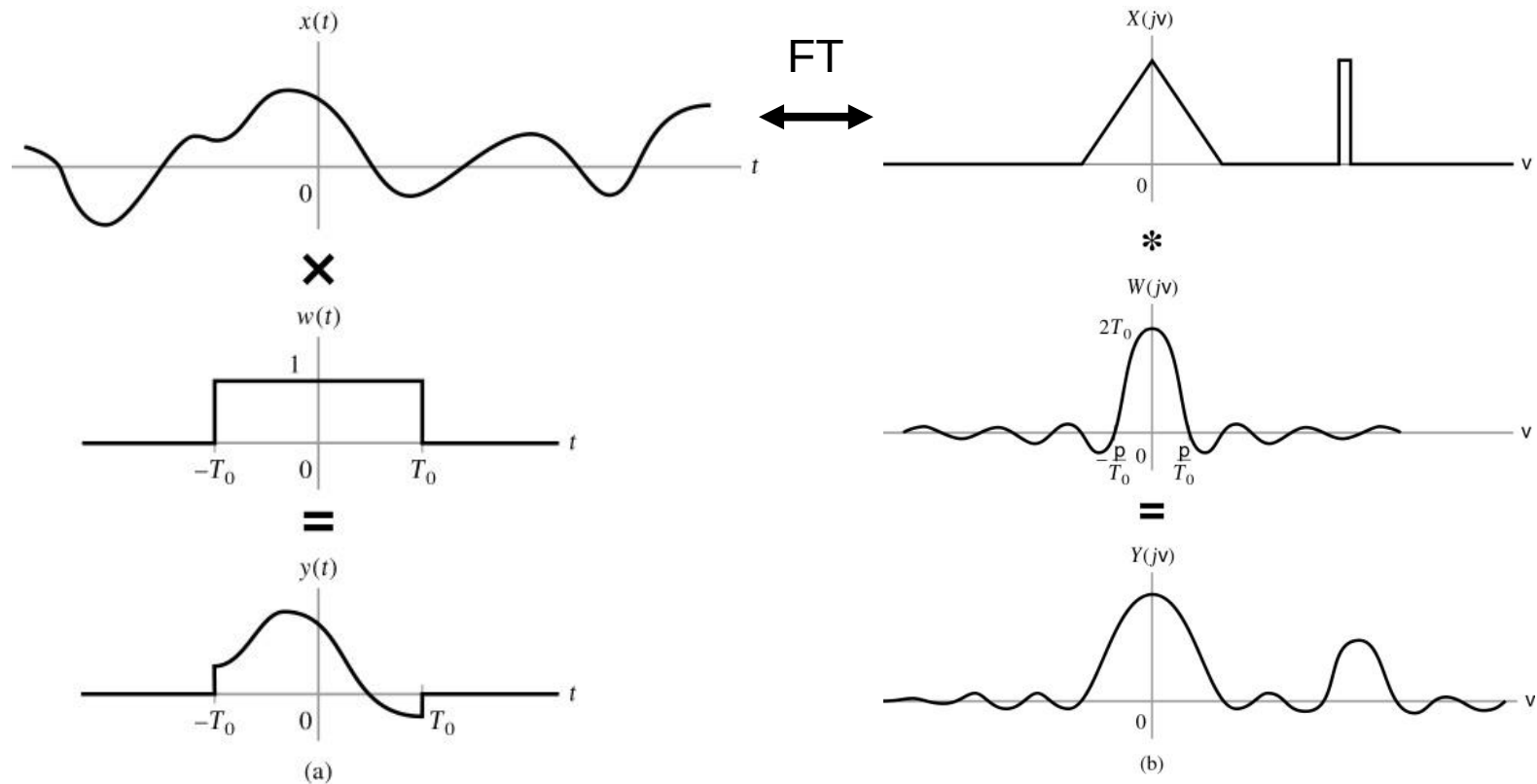
where,

$$W(j\omega) = \frac{2}{\omega} \sin(\omega T)$$



3.14 Multiplication Property

- The effect of windowing



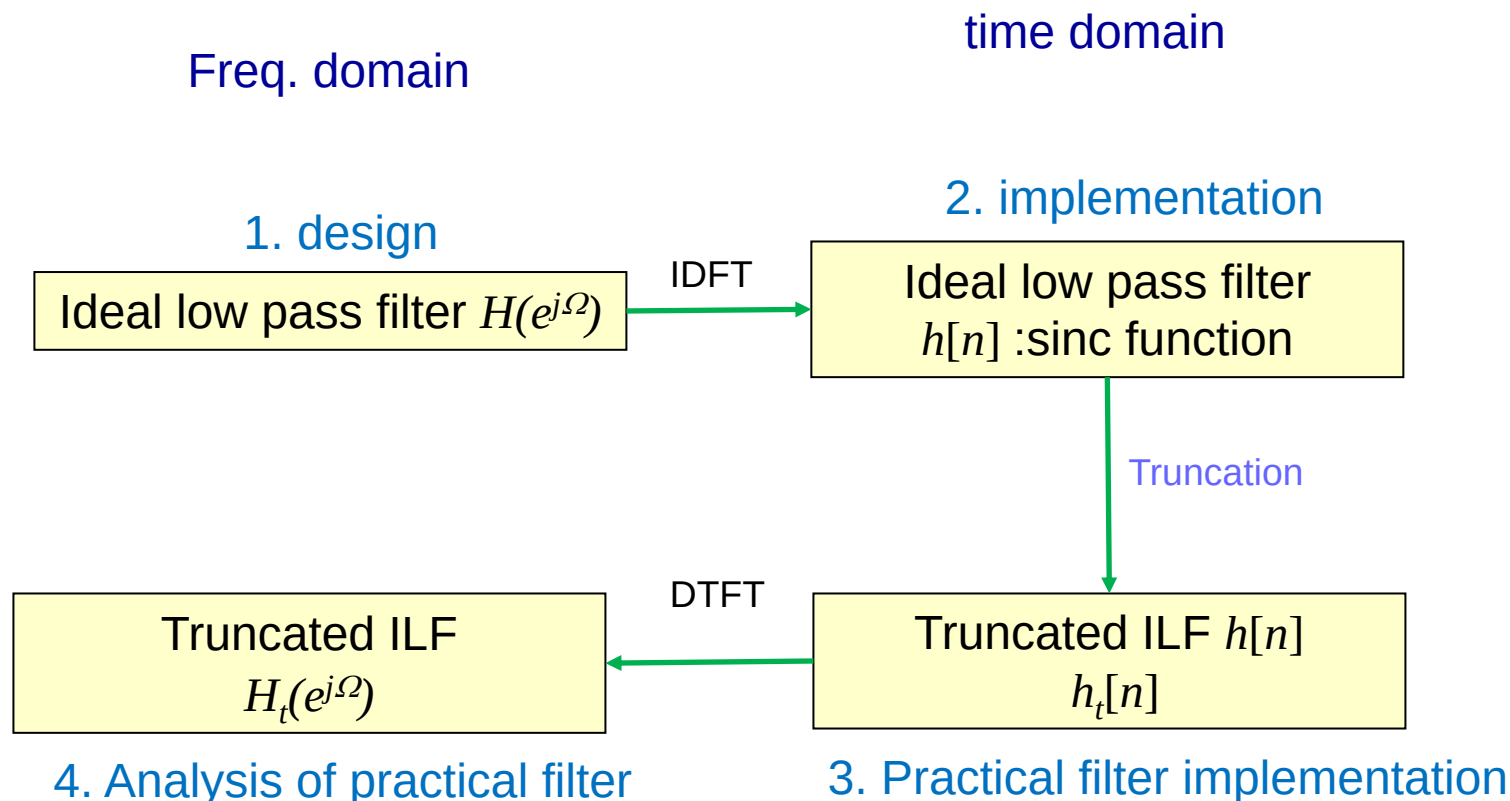
Time domain

Frequency domain



3.14 Multiplication Property

Ex) Truncating the Impulse Response
- ideal LPF

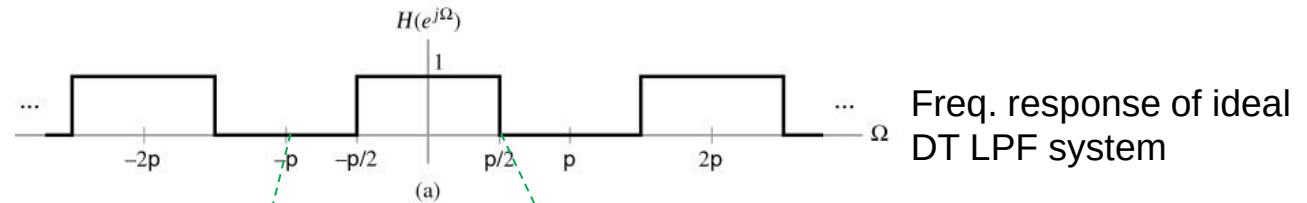


3.14 Multiplication Property

- The effect of truncating the impulse response of a discrete time system

- Design ideal LPF

$$H(e^{j\Omega}) \xleftrightarrow{DTFT} h(n) = \frac{1}{\pi n} \sin\left(\frac{\pi n}{2}\right)$$

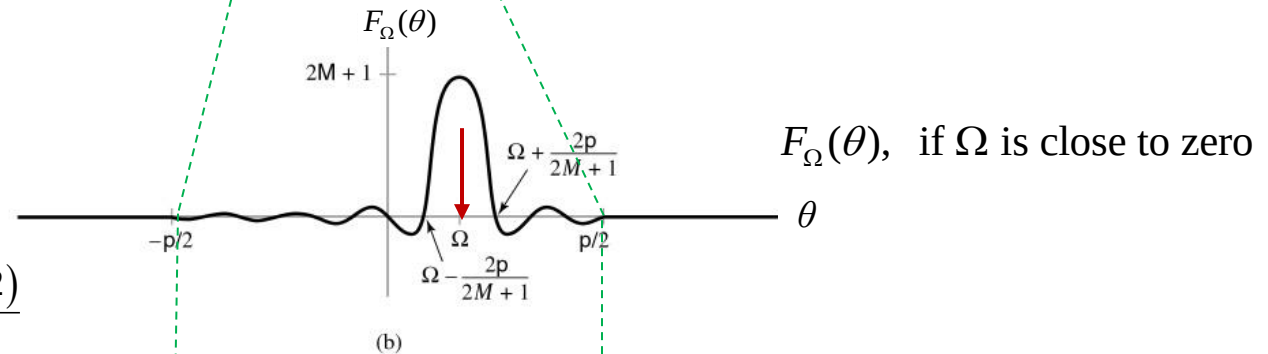


Freq. response of ideal DT LPF system

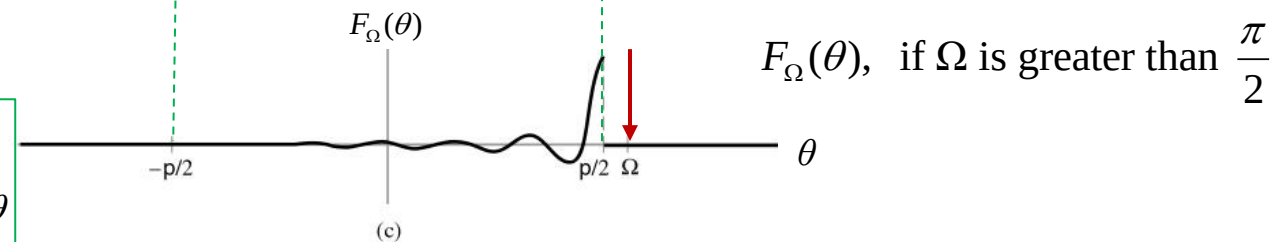
- Truncating

$$w[n] = \begin{cases} 1, & |n| \leq M \\ 0, & \text{otherwise} \end{cases}$$

$$\Rightarrow W(e^{j(\Omega-\theta)}) = \frac{\sin((\Omega-\theta)(2M+1)/2)}{\sin((\Omega-\theta)/2)}$$



$F_{\Omega}(\theta)$, if Ω is close to zero

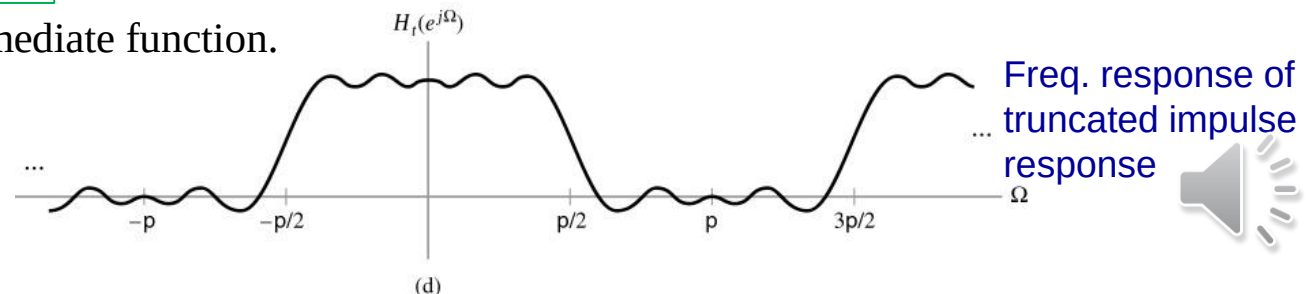


$F_{\Omega}(\theta)$, if Ω is greater than $\frac{\pi}{2}$

- Analysis

$$H_T(e^{j\Omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\theta}) W(e^{j(\Omega-\theta)}) d\theta$$

For convolution, θ is an intermediate function.



Freq. response of truncated impulse response



3.15 Scaling Property

▪ Scaling of time variable

→ Fourier transform $z(t)=x(at)$

$$\begin{aligned} Z(j\omega) &= \int_{-\infty}^{\infty} z(t)e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(at)e^{-j\omega t} dt \end{aligned}$$

Let $\tau = at$,

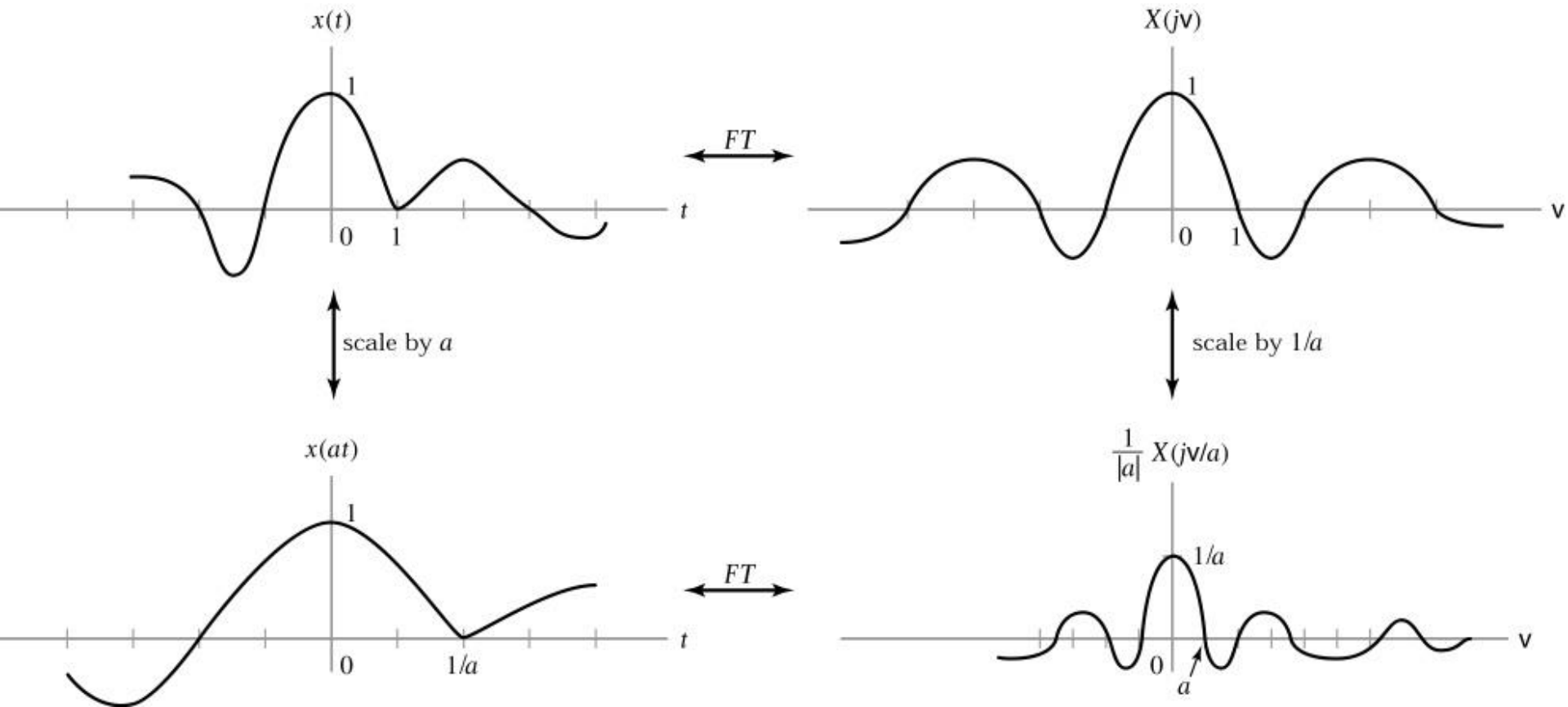
$$Z(j\omega) = \begin{cases} \frac{1}{a} \int_{-\infty}^{\infty} x(\tau)e^{-j(\omega/a)\tau} d\tau, & a > 0 \\ \frac{1}{a} \int_{\infty}^{-\infty} x(\tau)e^{-j(\omega/a)\tau} d\tau, & a < 0 \end{cases}$$

$$Z(j\omega) = \frac{1}{|a|} \int_{-\infty}^{\infty} x(\tau)e^{-j(\omega/a)\tau} d\tau$$

$$z(t) = x(at) \overset{FT}{\longleftrightarrow} \frac{1}{|a|} Z\left(\frac{j\omega}{a}\right)$$



3.15 Scaling Property



The FT scaling property. The figure assumes that $0 < a < 1$.



3.15 Scaling Property

▪ Differentiation and Integration

Ex) IFT? use of multiple Fourier representation properties.

$$X(j\omega) = j \frac{d}{d\omega} \left\{ \frac{e^{j2\omega}}{1 + j(\omega/3)} \right\}$$

(sol)

$$s(t) = e^{-t}u(t) \xleftrightarrow{FT} S(j\omega) = \frac{1}{1 + j\omega}$$

$$\text{then, } X(j\omega) = j \frac{d}{d\omega} \left\{ e^{j2\omega} S\left(j\frac{\omega}{3}\right) \right\}$$

$$\left(-jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega} X(j\omega) \right)$$

$$\text{Define } Y(j\omega) = S\left(\frac{j\omega}{3}\right)$$

$$y(t) = 3s(3t) = 3e^{-3t}u(3t) = 3e^{-3t}u(t)$$

$$\text{Define } W(j\omega) = e^{j2\omega}Y(j\omega)$$

$$w(t) = y(t+2) = 3e^{-3(t+2)}u(t+2)$$

$$X(j\omega) = j \frac{d}{d\omega} W(j\omega)$$

$$\therefore x(t) = tw(t) = 3te^{-3(t+2)}u(t+2)$$



3.16 Parseval Relationships

Parseval Relationships for the 4 Fourier Representation

Representation	Parseval Relationship
<i>FT</i>	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$
<i>FS</i>	$\frac{1}{T} \int_{\langle T \rangle} x(t) ^2 dt = \sum_{k=-\infty}^{\infty} X[k] ^2$
<i>DTFT</i>	$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(e^{j\Omega}) ^2 d\Omega$
<i>DTFS</i>	$\frac{1}{N} \sum_{n=\langle N \rangle} x[n] ^2 = \sum_{k=\langle N \rangle} X[k] ^2$



**THANK
YOU**

